

Water resources planning and management issues are rarely simple. Projects focused on addressing and finding solutions to these issues are also rarely simple. These projects too need to be planned and executed in ways that will maximize their likelihood of success, i.e., will lead to useful results. When decision-makers and other stakeholders disagree over what they want, and what they consider useful and helpful, the challenge facing project planners and managers is even more challenging. This chapter offers some suggestions on project planning and management. These suggestions reflect years of experiences the writers and their institutions, have had planning and participating in various water resources development projects, at various scales, in many river basins and watersheds throughout much of the world.

Each water resources system is unique, and the specific application of any planning and analysis approach needs address the particular issues of concern as well as adapt to the political environment in which decisions are made. What is important in all cases is that such planning and analyses activities are comprehensive, systematic and transparent, and are performed in full and constant collaboration with the region's planners, decision-makers, and the interested and affected public.

13.1 Water Management Challenges

Managing water is important. The effectiveness of strategies for dealing with water availability, quality, and variability is a major determinant of the survival of species, the functioning and resilience of ecosystems, the vitality of societies, and the strength of economies. Humans have been managing water and adapting to surpluses and shortfalls since the dawn of civilization, and especially since the early origins of agriculture. There is evidence across the globe of thousands of years of dam building and canal construction to direct water toward crops of various kinds. Though the tools and infrastructure water managers can use today are dramatically more sophisticated than those used in the past and the scale on which water managers work is much larger in almost all cases, the activities are still very much the same: managing floods and droughts through harvesting and storing water above or underground, delivering water across long distances through pipelines and canals, treating, distributing water supplies to where they are needed, collecting, and treating the resulting wastewaters all designed to meet a variety of economic, public health, environmental, and social objectives.

In regions witnessing increasing human populations demanding more energy and more food together with a more uncertain climate has led to a complicated dynamic interconnected web of physical, economic, and social components with many opportunities for intelligent adaptive management interventions. These interventions that change the distribution of water quantities and qualities over time and space can result in substantial economic, environmental, and social benefits. They can also introduce unexpected costs and risks. The constraints are physical (as with the large inputs of energy required for desalination), geographical (depending on the available suitable locations for reservoirs), financial (building, operating and maintaining infrastructure required to manage water is expensive), political (nobody wants to relinquish rights to scarce water without compensation), and ethical (what uses deserve to be prioritized, and how they relate to the needs of the environment).

Trade-offs are fundamental when allocating water to various sectors of society. Water is linked to the production of energy, food, industrial products and to human health and the condition of the broader environment. For many kinds of water uses, allocating water to one use usually means less water available for other uses. Consumptive use for agriculture, industry, or cities almost always involves trade-offs, as do mandates for instream flows to protect ecosystems or fisheries. But even consumptive uses do not diminish the total amount of global water. Consumption shifts water to a different part of the hydrological cycle: for example, from liquid to vapor, from clean to contaminated, or from fresh to salty.

Choices about managing water trade-offs involve more than hydrology and economics. They involve people's values, ethics, and priorities that have evolved and been embedded in societies over thousands of years. The juxtaposition of hydrology, economics, and values is at the crux of the water–climate–food–energy–environmental and society (people) nexus. While it

is unreasonable to think that models of water resource systems will or even should include each component of this interconnected interdependent nexus of components, analysts must be cognizant that the part of system that they model is interacting with and being influenced by those components assumed exogenous to the system.

13.2 Water Resources System Components, Functions, and Decisions

13.2.1 Components

For the purposes of planning and management water resource systems include three components:

- The *natural resource system* (NRS) component consists of the streams, rivers, lakes, and their embankments and bottoms, and the groundwater aquifers, and the water itself. This includes the abiotic or physical, biological, and chemical (“ABC”) components in and above the soil. It also includes the infrastructure needed to collect, store, treat, and transport water such as canals, reservoirs, dams, weirs, sluices, wells, pumping stations, pipes, sewers, and water and wastewater treatment plants, and the policies or rules for operating them.
- The *socioeconomic system* (SES) component is the water using and water-related human activities. This component can also include the stakeholders, i.e., the interested and affected public.
- The *administrative and institutional system* (AIS) component are the institutions that are responsible for the administration, legislation and regulation of the supply (NRS) and the demand (SES) components of the water resource system (WRS). This component includes those institutions that plan and build and operate the infrastructure required to insure that water is where and when and in the condition needed in ways beneficial to society.

13.2.2 Functions

Table 13.1 presents a framework of water resource system functions. This framework distinguishes between tangible and intangible functions. Tangible functions can be described quantitatively. For example, hydropower generation or municipal water supply, may be assigned a monetary value. Intangible functions are activities such as nature conservation or preserving a beautiful view that are hard to quantify in monetary terms. In between are environmental functions, some of which may be given quantitative values and others valued only indirectly, such as by using the opportunity cost associated with a particular target. The self-purification process of a river, for example, may be assigned a value by comparing this “work done by nature” with the costs of the least cost alternative that accomplishes the same results, such as constructing, maintaining and operating a wastewater collection and treatment system.

13.2.2.1 Subsistence Functions

Communities depend to a large extent on water for household uses, and for irrigating home gardens and community outdoor green and recreation areas. They may also use streams, paddy fields, ponds, and lakes for fishing. These uses are often neglected in national economic accounts, as they are not marketed or otherwise assigned a monetary value. However, if the WRS becomes unable to provide these products or services, this may well be considered an economic loss.

13.2.2.2 Commercial Functions

Commercial uses of water resources are reflected in national economic accounts because they are marketed or otherwise given a monetary value, e.g., the price to be paid for domestic water supplies. Catching fish for sale by individuals and commercial enterprises is an example. These uses have a commercial value and most are also consumptive in nature. The concept of

Table 13.1 Functions of the water resources system

functions	description	examples
subsistence functions	local communities make use of water and water-based products which are not marketed	◦ local drinking water supply ◦ traditional fishing ◦ subsistence irrigation
commercial functions	public or private enterprises make use of water or water-based products which are marketed or otherwise given a monetary value	◦ urban drinking water supply ◦ industrial water supply ◦ irrigation ◦ hydro-power generation ◦ commercial fishing ◦ transportation
environmental functions	regulation functions, non-consumptive use	◦ purification capacity ◦ prevention of salt intrusion ◦ recreation and tourism
ecological values	values of the WRS as an ecosystem	◦ integrity ◦ gene pool, bio-diversity ◦ nature conservation value

“nonconsumptive use” should be regarded with certain reservations. Nonconsumptive water use may alter the performance of the WRS in various ways. For example, consider reservoirs built for hydropower. Reduced sediment and fish passage and increased evaporation losses may impact downstream ecosystems and users. Second, operation of the reservoirs for the production of “peak power” may alter the flow regimes downstream, and this can adversely affect downstream ecological habitats and users. Finally, water quality problems resulting from reservoirs may impact users and ecosystems. Another example of partly nonconsumptive use is inland water transportation. Oil and chemical pollution caused by water transport activities can affect other users and the ecosystem that depend upon the water resources. Moreover, inland water transportation may involve a real consumptive demand for water. If water depths are to be maintained at a certain level for navigational purposes, releases from reservoirs may be required which provide no value to other water users. An example is the Lower Nile system, where water is released from Lake Nasser to enable navigation and energy generation during the so-called winter closure. This water could otherwise remain stored for (consumptive) use by agriculture during the growing season.

13.2.2.3 Environmental Functions

The drainage basin of a river fulfills a series of environmental functions that require no human intervention, and thus have no need of regulatory systems. These functions include self-purification of the water and recreational and tourism uses. It is sometimes difficult to assign values to environmental functions. They may be assessed by using opportunity costs, calculated as the costs of providing similar functions in other ways, e.g., the cost of additional wastewater treatment. Lower bounds on recreational and tourism values may be estimated by assessing the economic benefits accruing from the use of tourist facilities including hotels, and/or the revenue obtained from the sale of fishing licenses.

13.2.2.4 Ecological Functions

Rivers, streams, and lakes and their associated wetlands, floodplains, and marshes offer an environment for aquatic species. Land–water ecotones (transition areas between adjacent ecological communities) are known to harbor a rich assemblage of species, and are also important for the diversity of adjacent ecological communities. These ecological entities have an intrinsic ecological value irrespective of actual or potential human use. There are many concepts and expressions that describe this ecological value: “heritage value,” “aesthetic value,” “nature value,” “option value,” “existence value,” among others.

Box 13.1. Definitions

Policy goal: what do we want to accomplish?

Strategy: how do we want to do it?

Decision: what are we going to do?

Scenario: the external economic, environmental, or political situation affecting our strategy and decision.

13.2.3 Goals, Strategies, Decisions, and Scenarios

In planning projects the terms *goal*, *strategy*, *decision*, and *scenario* are frequently used. In popular use their distinction is often confusing. In this book we have used the following meanings:

- A *goal* defines what is to be achieved or how some target is to be met. Goals identify needs, prioritize issues and define targets and constraints on the actions to be taken to meet the targets. Goals may define preferred courses of action. For example, the goal might be to apply user-oriented demand management measures rather than relying on large-scale water supply infrastructure development.

- A *strategy* is defined as a logical combination of individual measures or decisions that accomplishes the stated goals and satisfies the constraints imposed on the WRS. For example, the construction of a reservoir plus the widening of the canal downstream and the increase of the intakes of the irrigation system all in an effort to reduce the risk of damage to the agriculture sector in a drought prone area is one strategy. An alternative strategy might be to implement a cropping pattern that uses less water.
- A *decision* is the implementation of a particular strategy or course of action. A distinction can be made between:
 - *Technical (structural) measures*: modifications of elements of the water resources infrastructure such as canals, pumping stations, reservoirs, and fish ladders. Technical measures often include managerial measures such as better ways of using the infrastructure.
 - *Ecological (nonstructural) measures* to improve the functioning of ecosystems, for example, by introducing fish fry in spawning areas, or large herbivores.
 - *Economical measures* to induce water consumers to alter their use of water by changing the price of the resource use (through charges, taxes, or subsidies).
 - *Regulatory measures* to alter the use of water (through land-use zoning, permits, pollution control and other forms of restrictive legislation).
 - *Institutional measures* specifying which governmental agencies are responsible for which functions of the WRS, and specifying the necessary interactions between the public and private sectors involved.
- A *scenario* is defined as the environment exogenous to the water system under consideration that cannot be controlled. Examples of scenario variables include rainfall and other aspects of the climate, demographical trends and changes, production functions (including crop water requirements), and most economic

variables relating to benefits and costs. What should be considered as a scenario and what as a decision variable may depend on the system boundaries that have been defined.

13.2.4 Systems Approaches to WRS Planning and Decision Making

Literature on the systems approach to planning often emphasizes the mathematical techniques used by practitioners of this approach. This book is no exception. Most of it is devoted to modeling water systems. The use of mathematical tools, however, is only part of what constitutes a systems approach. The approach applied to complex systems of many interdependent components, involves:

- building predictive models to explain system behavior,
- devising courses of action (strategies) that combine observations with the use of models and informed judgments,
- comparing the alternative courses of action available to decision-makers,
- communicating the results to the decision-makers in meaningful ways,
- recommending and making decisions based on the information provided during these exchanges between analysts, planners, and decision-makers and stakeholders, and
- monitoring and evaluating the results of the strategies implemented.

Systems analysis and policy analysis are often considered as being the same. If a distinction is to be made, one might define systems analysis as being applicable to more than just policy issues or problems. It can be applied to any system one wants to analyze for whatever reason. System diagrams or conceptual models identifying system components and their linkages are important tools in systems analysis. A system diagram

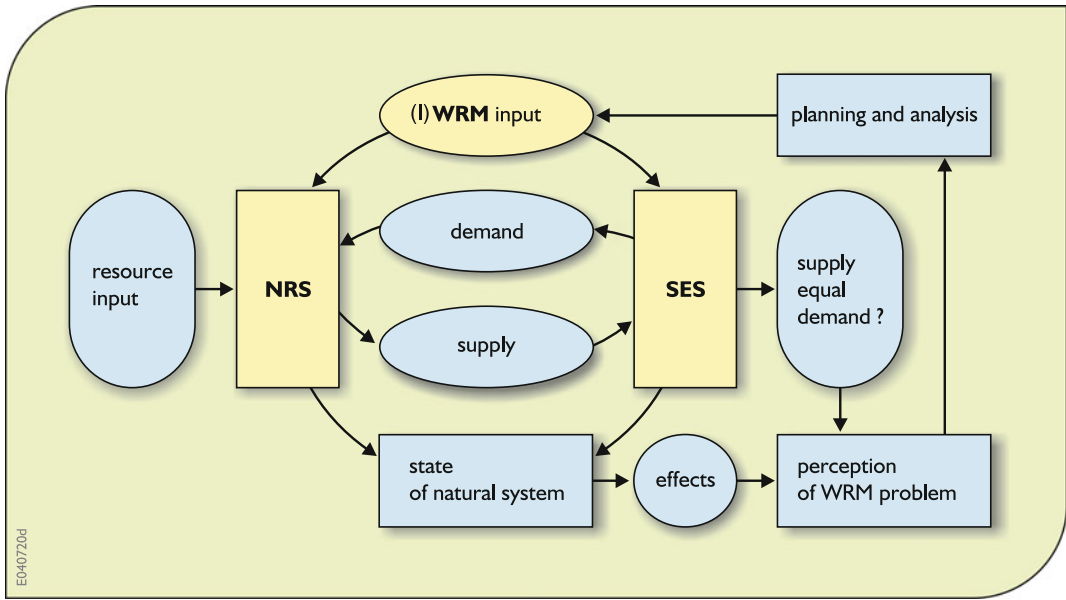


Fig. 13.1 Identification of a water resources management (WRM) problem

represents cause–effect relations among the components of the overall system. An example of the use of system diagrams in analyzing water resources problems is presented in Fig. 13.1.

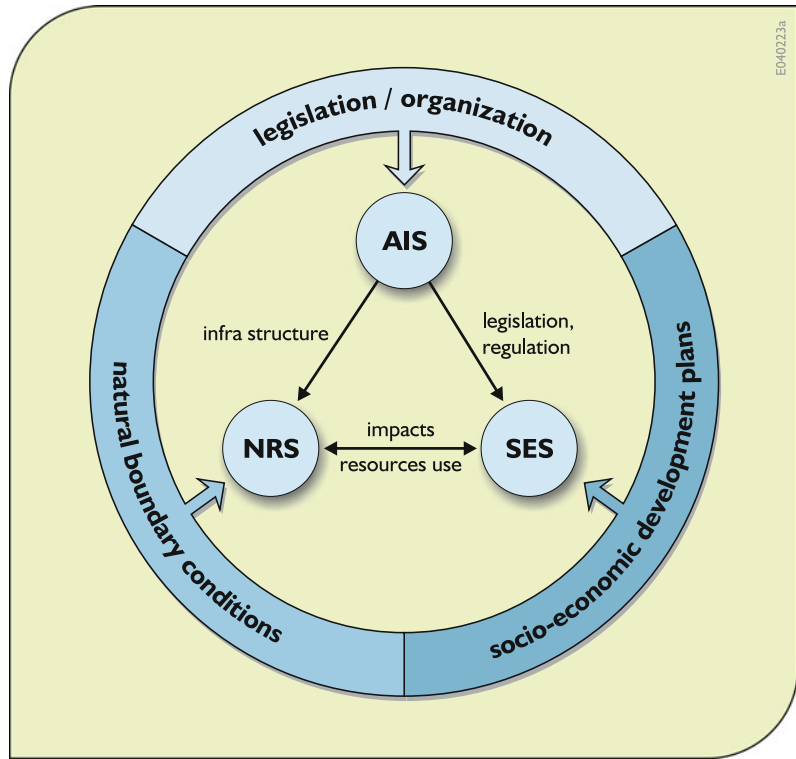
As Fig. 13.1 shows, water using activities may face two problems. First, the quantity demanded may be greater than the supply; second, they adversely impact the natural system (e.g., generate pollution or alter the water level). The perception of these problems can motivate analysis and planning activities, which in turn can result in management actions. The figure shows that the problems can be addressed in two ways: either by implementing demand-oriented measures (addressing the water use, i.e., SES), or by developing infrastructure that impacts the NRS). Demand-oriented measures aim to reduce water use and effluent discharge per unit of output. Supply-oriented measures on the other hand are aimed at increasing the water supply so that the magnitude and frequency of shortages are reduced or at increasing the assimilative capacity of the receiving water bodies. Which measure or combination of measures is most effective depends on the criteria selected by the implementing authority.

13.3 Conceptual Description of WRS

Water resources management aims to increase the benefits to society from the existence and use of water (NRS). Just how best to do it is society’s (SES) choice, commonly made through its governing institutions (AIS). These three “entities” are depicted in Fig. 13.2.

The management actions among the components of a WRS system are depicted by the arrows shown in Fig. 13.2. The arrows represent only the actions, not the information flows. There must be information feedbacks, otherwise effective management would be impossible. Each of the three systems is embedded within its own environment. The NRS is bounded by climate and physical conditions; the SES is formed by the demographic, social, and economic conditions of the surrounding economies; and the AIS is formed and bounded by the constitutional, legal, and political system it operates within. Boundary conditions are usually considered fixed, but in some cases they may not be. For example, climatic conditions

Fig. 13.2 Context for water resources planning involving the natural resource, socioeconomic and administrative–institutional systems



may be considered to be changing due to global warming. Similarly for laws and regulations. Whether and, if so, when to consider the possibility of changes in this “external” environment should be decided at the start of any planning project.

Consider, for example, regional economic . This predicted growth is often treated as given. If the water resources available cannot sustain this projected growth (or only at very high costs), it may be appropriate to reconsider this assumed growth. By learning the consequences of unrestricted growth at the regional level, planners can consider the desirability of other options that might be considered at higher (usually national) planning levels. This is represented in Fig. 13.2 by the border frame “socioeconomic development plans”. In fact, the arrow pointing inwards to the SES is reversed in such a case: the analysis provides information to a higher planning level that can change the boundary conditions.

13.3.1 Characteristics of the Natural Resources System

The natural resources system (NRS) is defined by its boundaries, its processes, and its control measures.

13.3.1.1 System Boundaries

The study area of a planning project will often coincide with an administrative boundary (state, county, district, province, etc.). However, a WRS is typically defined by its hydrologic boundary. These political and hydrologic boundaries can differ. Clearly, any planning project for a WRS must include the larger of these boundaries, but not necessarily everything within them depending on the purpose of the study and the particular WRS. The consideration of problem sheds that contain the components that impact water sheds is often useful.

For the purposes of modeling it has often proven useful to subdivide the NRS into smaller units with suitable boundaries. Examples are subdivisions into a groundwater and a surface water system, subdivision of a surface water system into catchments and sub-catchments, and subdivision of a groundwater system into different aquifers or aquifer components. The definition of (sub) systems and their boundaries should be done in such a way that the transport of water across area boundaries can be reasonably determined and modeled.

13.3.1.2 Physical, Chemical, and Biological Characteristics

The *physical processes* in an NRS are transport and storage within and between its subsystems. For the surface water system, a distinction is usually made between the infrastructure of rivers, canals, reservoirs, and regulating structures (the open channel network) and the catchments draining to the open channel network. The *biological* and *chemical characteristics* define the biological and chemical composition of groundwater and surface water and the transport, degradation and adsorption processes that may influence this composition. The level of detail to which these characteristics are considered will depend on the requirements and threats they impose on the water using and water-based activities.

13.3.1.3 Control Measures

By adding or changing the values of system parameters defining design and operating policy options of NRS, water resources managers can change the state of the system. An example is the rule curve defining how much water to release and when for different purposes. Another example is the flow capacity of feeder canals. Increasing the capacity of these canals permits greater allocations of water to farmers. An example of nonphysical control that changes the state of the biotic system is the release of predator fish in reservoirs to reach a desired balance of species in the ecosystem.

13.3.2 Characteristics of the Socioeconomic System

Like the NRS, the SES has its boundaries, processes, and control measures.

13.3.2.1 System Boundaries

The economic and social system generally does not have a physical boundary like that of the natural system. Economic and social activities in a river basin, for example, are connected to the world outside that basin through the exchange of goods, people, and services. The factors that determine the socioeconomic activities to include in a project planning exercise will depend on the context of the problems and development opportunities being considered. Outside the boundary of the socioeconomic system are factors or conditions that are beyond the control of the WRS decision-makers.

13.3.2.2 System Elements and Parameters

The socioeconomic part of the WRS can be defined by identifying the main water using and water-related activities, the expected changes and developments in the study area, and the parameters whose values define these changes and developments. Examples of activities or economic sectors that may be relevant and of the type of information that has to be obtained to be able to describe the SES include:

- Agriculture and fisheries: present practice, location and area of irrigated agriculture, desired and potential developments, water use efficiency, and so on.
- Power production: existing and planned reservoirs and power stations, operation and capacity, future demands for electric energy.
- Public water supply: location of centers of population and industrial activities, expected growth, alternative resources.
- Recreation: nature and location, expected and desired development, water quality conditions.

- Navigation: water depths in relevant parts of the open channel system.
- Nature conservation: location of valuable and vulnerable areas and their dependence on water quality and quantity regimes.

Some examples of important system parameters of the SES are labor force and wage rates, price levels in relation to national and international markets, subsidies, efficiency of production and water use, and income distribution.

When identifying and analyzing activities in the study area, it is important to consider possible discrepancies between the opinions of individual actors or stakeholders and their representatives. For example, individual farmers may have different interests than suggested by the official agricultural organizations.

13.3.2.3 Control Measures

The functioning of the SES can be influenced by legislative and regulatory measures, and the price of water may be a particularly important factor in deciding how much is demanded. This price can be influenced by the water resources managers and used as a control variable. When the cost of water use represents only a small portion of the total cost of an activity, however, an increase in its price may have little if any impact on water use. In some cases water use is a necessity of life no matter how high the costs. In such cases, the price of water (or taxation for waste water discharges) may not be an acceptable control variable (except perhaps to inform stakeholders on the consequences of possible cost reduction measures).

13.3.3 Characteristics of the Administrative and Institutional System

The AIS, like the NRS and SES, has its boundaries (its authority or limits) and its processes including its ways of reorganizing for improved performance.

13.3.3.1 System Elements

Administrative and institutional settings vary with scale, and with the way governing institutions exist and operate. In many countries, but certainly not all, the institutional framework consists of:

- the central government, divided into sectors such as public works, irrigation, agriculture, forestry, environment, housing, industry, mining, and transport
- a coordinating body, for example, a national water board, to coordinate actions by various sectors of the national government
- regional bodies based upon the normal subdivisions of government, for example, provinces, districts, cities, and villages
- regional bodies based on a division according to the physical characteristics of the area, such as river basin authorities
- water user organizations, representing the interests of directly involved stakeholders, for example, in irrigation districts.

When initiating broad comprehensive water planning projects knowing the following information is useful:

- the ministries and coordinating bodies having authority and responsibilities related to water resources management
- the agencies involved in the preparation of water resources development plans
- existing national and regional water resources development plans and the authorities responsible for implementing these plans, establishing and enforcing regulations, and overseeing infrastructure construction and operation
- the existing legislation (laws and regulations) concerning water rights, allocation of water resources, water quality control, and the financial aspects of water resources management.

Other often useful information includes the policies and plans of various water-related sectors such as environment, agriculture, economy, transportation, urban development and energy.

13.3.3.2 Control Measures

From a systems point of view, the decision or control variables that can be changed in the AIS are less clear than in the case of the NRS and SES. Often measures can be taken to improve the functioning of the system, for example, by establishing coordinating bodies when these are not present, shifting responsibilities toward lower levels of government, privatization, and other measures. If they cannot be changed, at least possible beneficial changes can be identified and presented to those responsible for making decisions.

13.4 Framework for Analysis and Implementation

A water resources planning study generally comprises five general phases, as illustrated in Fig. 13.3. Although we do not suggest the use of any rigid framework, some distinct phases and activities can be recognized and used to structure the analysis as a logical sequence of steps. The description of these phases, the activities in them, and the interactions among the activities in them, is referred to as the analysis framework. A coherent set of models is typically used for the quantitative analyses aimed at identifying and evaluating alternative beneficial measures and strategies.

A decision process is not a simple linear sequence of steps as suggested in Fig. 13.3, but involves feedbacks to earlier steps. Part of the process is thus iterative. Feedback loops are needed when:

- solutions fail to meet current criteria
- new insights change the perception of the problem and its solutions
- essential system components and links have been overlooked
- goals and objectives or the scope of study change (e.g., due to changing political, international, developments in society).

Communication and interaction with the decision-makers are essential throughout the

duration of a planning project and the implementation of the selected development. To ignore this increases the risk of generating plans and policies that are no longer relevant or of interest to the client. Regular reporting (inception and interim reports, etc.) helps in effective communication, but a continuous dialogue is important throughout all stages or phases of the analysis.

Decision makers and stakeholders should be involved in each of the five (idealized) stages of this framework. Otherwise there is a risk of the planning project producing results that those potentially impacted will not support. Stakeholder involvement brings both knowledge and preferences to the planning process—a process that typically will need to find suitable compromises among all decision-makers and stakeholders if a consensus is to be reached.

The framework involves a series of decisions at the end of each stage. The divergence–convergence process for involving stakeholders in decision-making on the five analysis stages is illustrated in the rhombus approach of Fig. 13.4.

The first *inception stage* of the process identifies the subject of the analysis (what is to be analyzed and under what conditions), the objectives (the desired results of the analysis), and constraints (its limitations). On the basis of this analysis, during which intensive communication with the decision-makers is essential, an agreement on the approach for the remainder of the analysis needs to be achieved. The results of the inception stage can be presented in an inception report, which includes the work plan for the other phases of the analysis.

In the *situational analysis stage*, the tools for the analysis of the water resource system are selected or developed. Major activities in this phase typically include data collection and modeling. The models will be used to quantify the present and future problems in the system. Scenarios will be developed to describe the future boundary conditions for the system. Identifying and screening of alternative decisions can occur in this phase. If possible no regret measures will be identified for immediate implementation. A gradual improvement of the

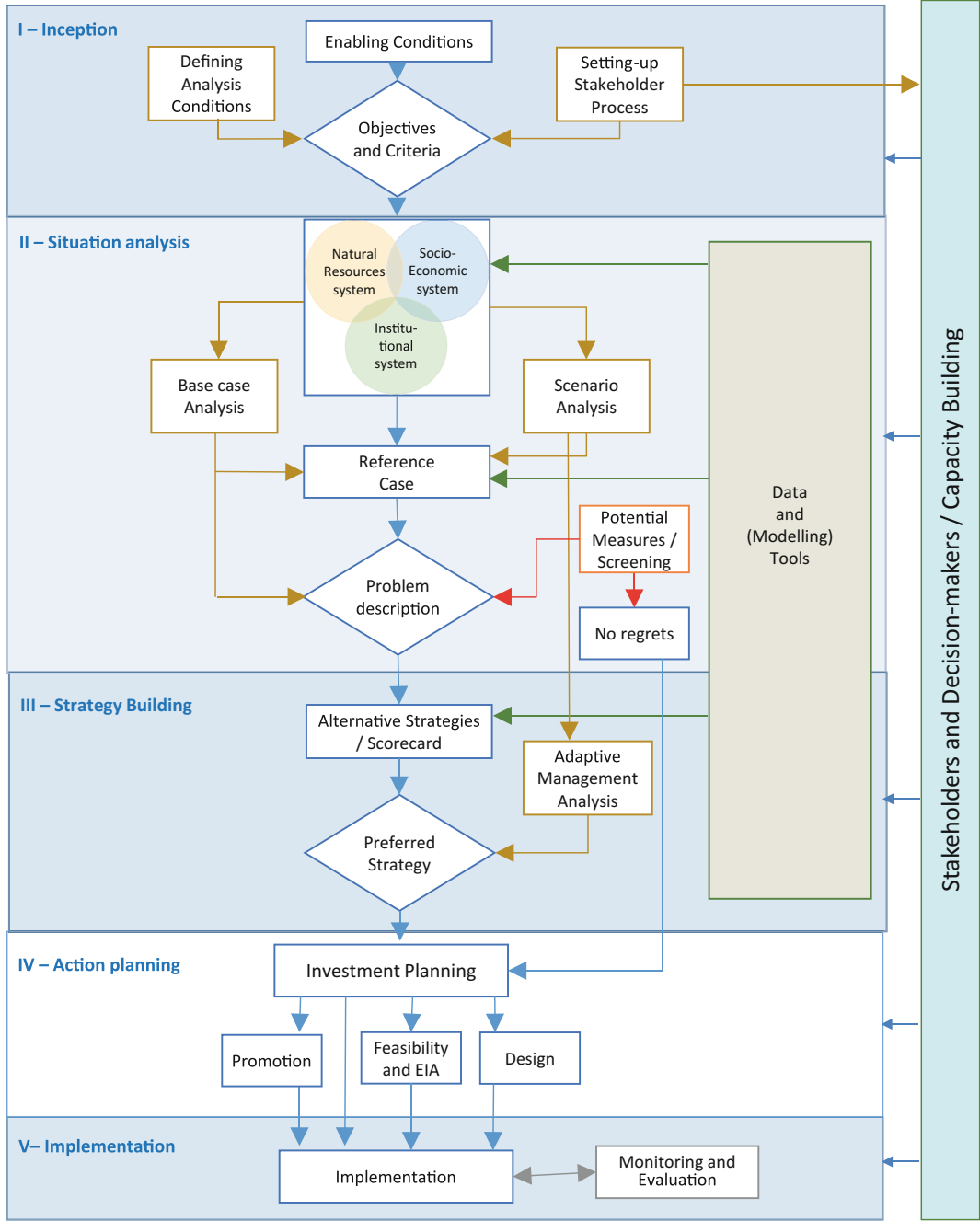


Fig. 13.3 Framework for analysis and implementation of water resources projects

understanding of various characteristics of the WRS is often obtained as the study progresses from limited data sets and simple tools to more detailed data and models. Interaction with the decision-makers will be greatly enhanced if they

or those they trust and communicate with are involved as part of the analysis team. More formal interaction can be structured through presentations of results in meetings and in interim progress reports.

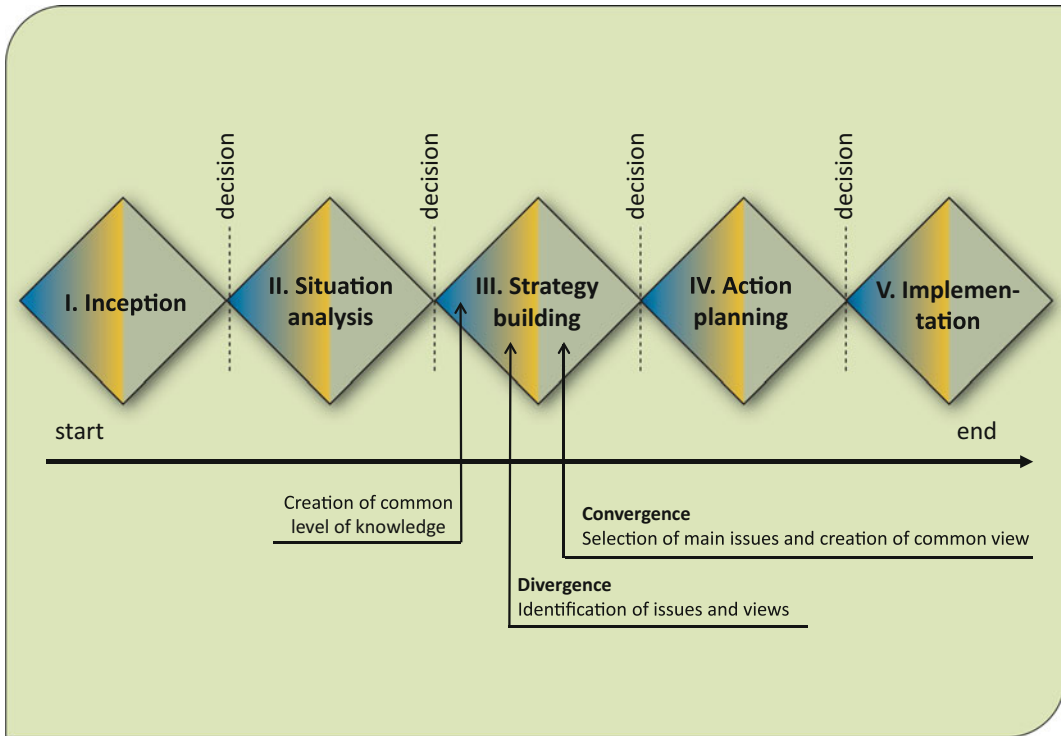


Fig. 13.4 Divergence—convergence process in decision-making

In the *strategy building stage* alternative strategies will be developed and discussed with the decision makers/stakeholders. This will include adaptive management elements to ensure that the preferred strategy is sufficient robust and flexible in case the future develops differently than expected.

In the *action planning stage* the selected strategy will be prepared for implementation. An implementation plan will be developed that describes what will be done, by who, how it will be financed, etc. This stage often requires also additional work on components of the strategy (such as feasibility and design studies), and environmental impact assessments (EIA). Promotion of the he selected strategy is needed to “sell” the proposed measures to public. Finally, institutional arrangements will have to be made to ensure a smooth implementation.

Finally, in the *implementation stage* the actual implementation will take place. Continuous monitoring and evaluation is needed to adjust the

implementation plan when this appears to be needed, for example, because the conditions (e.g., finances, social pressures, political mood) change.

Each stage or phase needs to provide the information desired by those institutions who will decide on what is best to do, and when, and how. What those governing institutions need to know to be better informed before making their decisions will of course vary among different planning projects. But whatever that information is the purpose of performing analyses is to create and communicate it. The results of the analyses performed in a planning project should be of no surprise to those reading them in a final project report. Again, communication between the project and the requesting institutions, and the affected public—the stakeholders—is essential throughout the project. This communication may not guarantee a consensus but it can certainly help the project team in their efforts to find it.

13.4.1 Step I—Inception Phase

Water resources planning studies are often triggered by specific management problems such as the need to increase power production or water supply reliability, the occurrence of droughts or floods, or the threat of water quality deterioration. The need for water resources planning in relation to other sector planning efforts may also be a trigger. Which parts of the WRS are studied and under what conditions follows primarily from the objectives of the study (and from the available budget, data, and time). The initiators of the study generally have more or less concrete ideas about the objectives and purpose of the analysis. However, these can change during a study.

The client's ideas about the problems and issues to be addressed will usually be described in a Project Formulation Document (PFD) or Terms of Reference (ToR). The very first activity of the project is to review and discuss the contents of these documents. If the subject (what needs analyzing) and objectives (what is to be accomplished) are adequately described in the ToR, the next step of the study is to specify and agree on the approach (how).

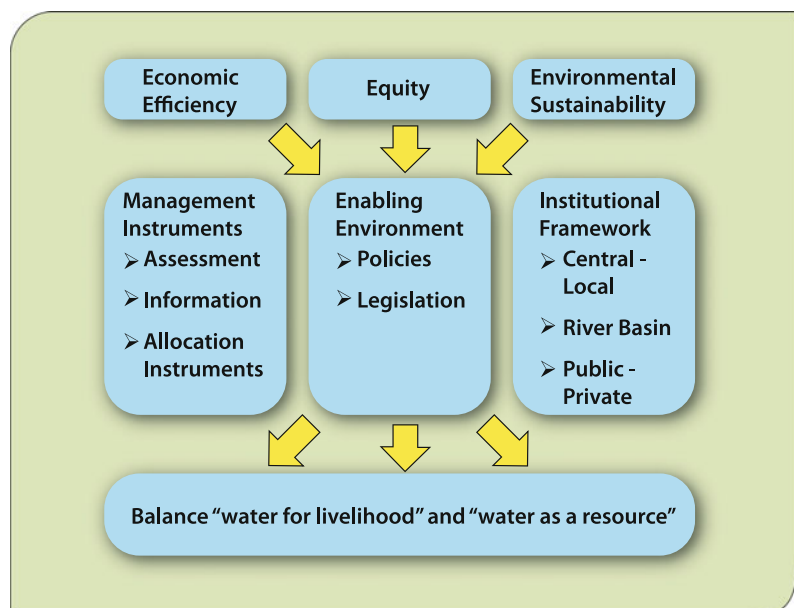
In many situations, however, the next task of the project will be to assist the decision-makers in further specifying the objectives and subject of

the analysis. For this activity, intensive communication is required with authorities involved in water resources planning and the stakeholders. They can provide information on the requirements of various interest groups related to water and on expected problems. It is not uncommon to have the stated objectives of a study differ from the actual (often unstated) objectives of the client (including just stalling for time hoping stakeholders will lose interest in a particular issue). Furthermore, objectives can change over time. As emphasized above, constant and effective communication between analysts and their clients is absolutely essential to the success of any planning project. We mention this often as it is not always easy given busy time schedules and often having to learn the differences in the meanings of various words or expressions (jargon) used by all parties.

13.4.1.1 The Enabling Conditions

In order to successfully carry out a good planning study certain conditions should be met. Most of these conditions are external to the project activities. This means that they should have been set before the planning exercise starts. A generic description of the enabling conditions for integrated planning is given in Background Paper no. 4 (GWP 2000) and is illustrated in Fig. 13.5.

Fig. 13.5 Enabling conditions (the “pillars”) for IWRM



- Enabling environment at national level:
 - national water legislation and national policies that guide the planning process and enables enforcement.
- Institutional framework:
 - existence of water institutions at national and regional level with qualified staff;
 - in case of river basin studies, existence of some kind of river basin organization (RBO) at river basin level.
- Management instruments:
 - availability of data, information, and tools that enables informed decision making.

In the Inception stage it should be determined which conditions are relevant for the specific planning exercise. This depends on the issues involved. If needed, institutional measures can be part of the planning project.

13.4.1.2 Setting Up the Stakeholder Involvement Process

The very first step is to set up the stakeholder involvement process. Which stakeholders to involve and how will depend on the specific basin and the issues to be addressed. In general two categories of stakeholders can be identified:

- the people and organizations that will be affected by the plan; and
- the people and organization that are needed to implement the plan.

In some cases a stakeholder analysis might be needed to determine the best stakeholder involvement process. More detail on involving stakeholders is given in Sect. 13.5.1.

13.4.1.3 Defining Analysis Conditions

In addition to the more legal and institutional oriented conditions as described in Sect. 13.4.1.1 it is necessary to get agreement on the analysis conditions for the planning study. This includes:

- The base year for the study:
 - the most recent year for which basic data on the present situation is available;

- The time horizon(s) for the study:
 - this may include short term (e.g., 5 years), medium term (e.g., 20 years) and long term (>25 years);
- The discount rate to be applied in the economic analysis:
 - taken as specified by (e.g.) the Ministry of Finance or Economic Affairs, or by the financier of the planned investments (e.g., ADB, World Bank and JICA);
- System boundaries of NRS, SES, and AIS—the components and the level of detail that will be included:
 - e.g., will the coastal zone be included in a river basin study?
 - are the results to be presented at local government unit level?
- Time periods based on within- and over-year variability of systems processes and inputs
- Scenario assumptions concerning factors external to the WRS, such as the growth of population, food and energy consumption and prices. See also Sect. 13.4.2.4.
- System assumptions. These concern factors internal to the WRS, such as the response of crop production to improved cultivation practices, or the effectiveness of price incentives on per capita water consumption. These system assumptions can be subject of additional (sensitivity) analysis.
- Data, time, and budget constraints. Studies have to be executed within constraints of available data, time, and budget.

The choice of the time horizon is often given insufficient attention. Official planning horizons (e.g., 5, 10, and 25 years) are typically used as time horizons for elements of the analysis. However, one should also consider the time-scales of the system and the processes within it. System components will have characteristic time scales. For example:

- Economic activities have life cycles that are usually determined by the amortization period of the investments. Time horizons of planning processes can be based on these conditions.

- Social institutions have time horizons that depend on the pace of legal/institutional and political decision making.
- Physical–chemical systems have time scales that depend on the response or restoration times of the systems. Restoration of polluted rivers, for example, may be achieved within a few months, while the restoration of a polluted groundwater aquifer may take decades.
- Ecosystems may have a time scale of a few weeks (algae blooms) or tens of years (degradation of mangrove forests), depending on the type of process or intervention.

To study the sustainability and ecological integrity of the resource system, time horizons should be tuned to the response times of the system rather than to a planning horizon only. Although more attention is now paid to sustainability, no operational procedure has been generally adopted to properly consider long-term effects in the evaluation process. Decision-makers tend to focus on short-range decisions even if they impose possible risks in the long term, because their political time horizons are often limited to (or renewable in) short terms and hence they prefer short-term political gains.

13.4.1.4 Objectives and Criteria

An essential activity in the inception phase is the translation of general objectives, as described in the ToR or in policy documents, into operational objectives that can be quantified. Examples of objectives and criteria are discussed throughout this book and especially in Chap. 9. The objectives and criteria used in a water resources management study in West Java, Indonesia are presented as an illustration at the end of this chapter.

National and regional development objectives

An essential component of an integrated plan is the connection of the plan and its objective to national development goals as well as to common international goals (e.g., the Sustainable Development Goals—SDGs). The plan should

refer to national policy priorities and indicate the contribution the plan will make to the various development goals. Required information is usually described in various national policy documents. In addition to the national policy documents any existing regional/provincial policy documents need to be taken into account. Each plan need to have an agreed objective that not only focuses on the main, but also expresses the relation with above mentioned national and other sector plans, as well as the contribution the basin can make in realizing these higher level plans.

Operational objectives, criteria and targets

If needed, the general objectives as stated in the national policy documents have to be translated into operational objectives for the specific area under consideration, e.g., a river basin. This should be done by specifying them in socioeconomic terms, amongst others, which are meaningful to the decision makers and stakeholders. For each objective evaluation criteria should be defined as a measure of how far the defined objectives have been achieved and, if possible, clear targets should be specified. Monitoring will indicate how far the objectives have actually been achieved. This process, illustrated in Fig. 13.6, is discussed in more detail in Chap. 9.

The evaluation criteria need to be comprehensive (i.e., sufficiently indicative of the degree to which the objective is achieved) and measurable. The criteria do not all have to be expressed in a single measurement scale. Criteria can be expressed in monetary and nonmonetary terms.

It may be useful to incorporate sustainability as an objective, and if so, it may also be useful to relate them to the UN Sustainable Development Goals (SDGs), the SDG targets and the indicators, that have been selected to monitor the SDGs.

Illustrative river basin case

Table 13.2 presents a scorecard that summarizes results of an analysis for a river basin case. The results of the Inception step (i.e., the objectives and criteria), for this river basin are given in the

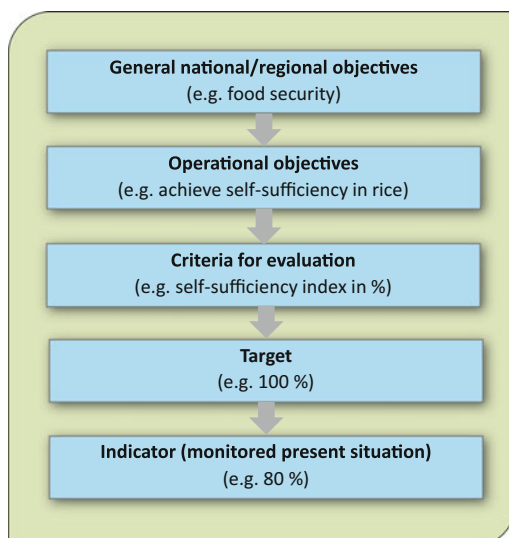


Fig. 13.6 Making objectives operational

first two columns of the table. They show that for this case five objectives were formulated. For each objective 2 or 3 criteria were identified that expresses in how far the objective is or will be achieved:

- Objective 1: Provide safe water and sanitation for the people;
 - % people access to safe drinking water;
 - % people access to sanitation facilities;
- Objective 2: Increase food production;
 - Irrigation area (ha);
 - Number of animal water points (#);
- Objective 3: Support economic sectors—industry and energy;
 - Water supplied to mining (% of demand);
 - Water supplied to industry (% of demand);
 - Hydropower generated (MWh);
- Objective 4: Protect the Environment;
 - Protected watershed area (km²);
 - Number of springs/sources protected (#);
 - Average class water quality rivers (class A to D);
- Objective 5: Decrease vulnerability to floods and droughts;
 - Vulnerability to floods—average damage (\$/year);
 - Vulnerability to droughts—average damage (\$/year).
- In addition two implementation-related criteria were formulated to evaluate the strategies:

Table 13.2 Example of a scorecard showing objective values associated with various strategies

Objectives and criteria		Base Year 2010	Targets		Perfect	Alternative (investment) strategies					
						Ref. case (no action)		Strategy 1		Strategy 2	
unit		2020	2030			2020	2030	2020	2030	2020	2030
Obj.1: Water and Sanitation											
% people access to safe drinking water	%	50%	63%	73%	100%	63%	73%	63%	73%	63%	73%
% people access to sanitation facilities	%	30%	50%	70%	100%	50%	70%	50%	70%	50%	70%
Obj.2: Food production											
Irrigation area	1000 ha	24	30	35	40	26	28	28	31	30	35
# animal water points	#	300	500	900	1000	400	700	500	900	500	900
Obj.3: Industry and Energy											
Water supplied to mining	%	30%	80%	90%	100%	40%	50%	50%	70%	80%	90%
Water supplied to industry	%	70%	80%	90%	100%	70%	70%	80%	90%	80%	90%
Hydropower generated	MWh	34	80	120	120	34	34	70	110	80	120
Obj.4: Environment											
Protected watershed area	km ²	1200	2500	3500	3500	2000	2500	2500	3000	2500	3500
Number of springs/sources protected	#	300	600	900	900	400	600	500	700	600	850
Average class water quality rivers	I - V	II	III	IV	V	II	III	III	III	III	IV
Obj.5: Vulnerability											
Vulnerability to floods - average damage	m€/yr	120	< 78	< 50	0	100	80	100	80	78	50
Vulnerability to droughts - average damage	m€/yr	200	< 50	< 30	0	160	120	80	40	50	30
Implementation information											
Required investments	m€		-	-	-			400	650	600	1200
B/C ratio economic categories (Obj.2, Obj.3)	-		> 1,3	> 1,2	-			1.3	1.2	1.2	1.1

- Required investments (\$);
- Benefit/Cost ratios of economic categories (</>).

13.4.1.5 Work Plan and Decision-Making

Once it is clear “what” will, as well as what will not, be analyzed and “why”, analysts can specify “how” this will be done. A description of the system to be analyzed includes the conditions and the assumptions under which the analysis will be performed.

All required activities can be combined in a work plan. It is often advantageous to develop a critical path network of the various analysis tasks. Critical path networks define the sequence of various tasks required to complete an analysis, or indeed the entire planning project, and their start and finish times. This will guide the allocation of personnel and identify the time needed to perform such tasks. These networks can be updated as the project proceeds. Such networks are useful for scheduling activities and personnel involved in the project, and for ensuring (or at least increasing the probability) that data and personnel will be available for each activity when needed and when decision-makers and stakeholders are to be involved in the analyses or in workshops or meetings focused on improved understanding of project progress and goals.

Data Availability

An important boundary condition for studies is often the availability of data and other information required for the study. The availability of data determines the level of detail and accuracy that can be achieved in the analysis. If few data are available, a more qualitative analysis may have to be performed. The required level of detail will primarily depend on the problems to be addressed and the objectives to be satisfied.

Level of detail One of the main tasks of a project leader is to motivate and manage the experts from various disciplines. Not staying focused on the appropriate level of detail is one of the most common causes for project failure. If the needed level of detail is underestimated at the

start of the project, the study will have to obtain the additional detail needed to fulfill the objectives of the analyses. Sometimes the right level of detail is chosen, but team members may be tempted to spend too much time addressing more detailed questions of interest to them and fail to come up with the information desired within the available time. Maintaining the proper level of detail is one of the main reasons for feedback loops in the analysis process.

Computational Requirements

An important element of the work plan will be the determination of the computational resources needed for the analysis. This includes mathematical models, databases, GIS, and the like. Together these must be used in a way that describes the system and permits an evaluation of possible measures and strategies under different scenarios at the level of detail desired. Often a combination of simulation and optimization models has proven useful.

For the purposes of analysis, the study area is typically subdivided over space and time into smaller units considered to be homogeneous with respect to their characteristic parameters. Each unit can be included in mathematical model(s). The number of elements required for the analysis depends on the issues being addressed, the complexity of the study area, the measures to be studied and the availability of data. It generally is wise to start with a preliminary schematization with the minimum number of elements. If more spatial or temporal detail is required model elements can be subdivided. The assumptions and conditions under which analyses are undertaken should be specified in close cooperation with those institutions overseeing and contributing to the study.

Work Plan

The results of the inception phase are documented in an inception report. This report can serve as a reference during the execution of the study. An essential part of the report is the proposed work plan, in which time, budget and human resource allocations to various activities

are specified. This work plan typically includes bar charts (possibly derived from critical path analyses) for activities and staffing, time schedules for deliverables, milestones, reporting procedures and similar features. The report should include a communication plan that describes the interaction between the decision-makers and stakeholders and the analysis team.

Inception Report

An inception report is a specific and concrete result of the inception phase. It contains the findings of and decisions made during the inception phase. It should make clear what will be studied, and why and how. In many cases it will also specify what will not be studied and why. The content of the inception report follows the subjects mentioned above. It is an important product because it contains all that has been learned in this first inception phase and that has been agreed upon between the analyst and the “client” (the decision-makers and the stakeholders).

A possibly even more important result of the inception phase, however, is the interaction between the analyst and the client that took place during this phase. It should state the client’s views about problems, objectives and other aspects. Project analysts must understand the client’s concerns, problems and objectives. Clients should feel they “own” the results of the inception phase and view the inception report as their own product, not merely a report of the planners, analysts or consultants. To achieve such ownership, frequent interaction must have taken place among the analysts, the decision-makers and stakeholders, to a much greater extent than is indicated in Fig. 13.3. This can be done in specific workshops, such as those devoted to the problem statement or to the specification of objectives and criteria.

13.4.2 Step II—Situation Analysis

In the situation analysis phase the study starts to dig deeper in the water resource system. Its various components will be studied in detail, data

will be collected and where necessary and possible the system components will be captioned in models. As much as possible this should be done in close collaboration with the stakeholders to ensure that the analysts and stakeholders have the same understanding of the system. Once these models are available a structured analysis can be carried out to quantify the present and future problems and a start can be made with identifying measures to address these problems.

13.4.2.1 Understanding and Describing the Water Resources System

A WRS comprises:

- Natural (Resources) System (NRS);
- Socioeconomic System (SES); and
- Administrative and Institutional System (AIS).

Each of the three systems is embedded within its own environment. The Natural Resources System is bounded by climate and (geo)physical conditions. The SES is formed by the demographic, social and economic conditions of the surrounding economies. The AIS is formed and bounded by the constitutional, legal and political system. The interlinkages of the three systems are illustrated in Fig. 13.7.

It is important that the plan includes a good description of the integrated elements of the

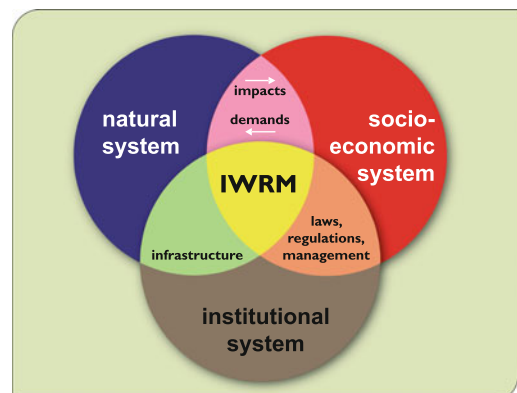


Fig. 13.7 Systems components of a WRS

WRS. Most decision-makers and stakeholders will be nontechnical or only know about a limited part of the overall system. To be able to make balanced decisions they should understand how the overall system functions and how interventions in one part of the system will impact other systems elements.

The situational analysis starts with an inventory of the *characteristics* of the WRS. This requires the reduction of a complex reality into a comprehensible description of system components and linkages. Choices have to be made about what (the detail that) should be included and what can be ignored. Such choices require engineering and economic judgment in combination with an understanding of the problems and possible measures that can be taken to improve system performance. The next step will be an inventory of the *activities* and *ongoing developments* that will determine how the system will perform in the future and what kind of additional activities can be expected. This can include autonomous developments (such as population and urban growth) as well as policy decisions that have been or may be taken that could influence the characteristics and performance of the WRS. An inventory of *policies and institutions* is helpful for identifying who is involved in the management and development of the system (and hence who should be involved in the analyses) and their objectives and opinions. This knowledge will contribute to the development of scenarios for the analyses.

Analysis of the Natural Resources System (NRS)

The NRS comprises the natural and engineered infrastructure, including the hydrometeorological boundary conditions. Models can be used to simulate the processes of water distribution through the infrastructure, taking into account the storage of water and water withdrawals to satisfy the demands of water-using activities. Such models have been introduced in many of the previous chapters of this book.

The results of the water quantity modeling may be the inputs for water quality models. The analysis of chemical components in the water

system is used to study the influence they have on the user functions or the biological system. The components and processes that are to be considered in the analysis should have been selected in the inception phase. The analysis of the biological system aims to determine the response of the ecosystems to water resources management (see Chap. 10). Since often there is too little exact information on individual biotic components and their behavior under different hydrologic and chemical regimes, models of ecosystems typically depend on habitat parameters.

Analysis of the Socioeconomic System (SES)

Developments in the SES determine the way demands on the NRS may change. Conversely, the development of economic activities within the study area may depend on the availability of water. For example, good supplies of relatively cheap surface water may stimulate the development of irrigated agriculture, or attract industrial activities that require large quantities of water for their production processes. Another example is the development of water-based recreation activities adjacent to a reservoir. These SES developments in turn increase the water demands. Economists or planners may be able to estimate future levels of the activities dependent on water discharges and storage levels. These relations can be incorporated into water resource planning models.

The starting point for an analysis of the SES is an assessment of the present economic situation with respect to the water-related activities and the factors that determine these activities. Past trends can help provide information on factors that have been decisive in bringing about the present situation and that may give clues about the likely impacts of future developments. One's attention should be on the most important factors that determine relevant water-related activities rather than on analyses of the total economy. However, the difficulty in forecasting economic development is the uncertainty about which factors will be decisive for this development.

Part of the data needed to develop planning models is the relation between the economic

activities and their water use. Data are needed that define the type and amount of water used by various activities. Data are needed identify the following with respect to each identified activity:

- the amounts of water (quantity and quality) demanded and consumed during which periods of the year and at which locations
- the amounts of water discharged and the pollution loads during which periods of the year and at which locations
- the benefits to the user if these amounts are made available
- the damage to the user if these amounts are not available
- costs that can be recovered by having the user pay for the water and its influence (both at the intake and the discharge sites of his activity) on the water use pattern.

All these data should be able to contribute to the estimates of future water demands, consumption and wastewater discharges per unit of activity. As well as the level of activities and the resulting water demands, knowledge of the geographical location of water using activities (the pattern of activities) is necessary. If the pattern of activities is not expected to change, the analysis can be focused on the present situation in the study area. If new activities are expected to develop within the study area and their water use characteristics are unknown, it may be necessary to study the water use characteristics of similar activities in other regions.

The resulting water demand data need not always be considered as “given.” Water-use coefficients can be changed through measures such as water pricing that aim at reaching a socially preferred use pattern. Technological developments may result in less water use and pollution load per person or unit of product. If supplies and demands are matched before the effects of such incentives are analyzed then one may over estimate needed capacities, because the “given” demands may be lower if water users are confronted with the costs as well as the benefits of water use. This type of internal feedback should be considered in the study.

Future water demands are often dependent on future scenarios. A water demand scenario is a logical but assumed combination of basic SES parameters and their effects on water-related activities, including the resulting water demands. An understanding of the functioning of the SES developed through the assessment of past and present trends is often helpful when formulating a limited number of consistent scenarios. Box 13.2 is an example of one such scenario.

Box 13.2. Example demand scenario

The water demand in an agricultural area depends largely on the availability of land and the crops being irrigated. The demand for agricultural products, however, will develop in an autonomous way. If the availability of water resources in a region is limited, the autonomous development of the agriculture sector will be limited as well, and one would predict a small increase in agricultural water demand. If the demand for agricultural products increases considerably and self-sufficiency in food production is an objective, then the political pressure for agricultural development to meet this objective may be considerable. The water demand corresponding to this desired agricultural development could show the need for further development of the water resources in the region.

Analysis of the Administrative and Institutional System (AIS)

An analysis of the AIS is required to identify any legal or regulatory or institutional constraints on water resources management. Attention must be given to the interaction between various authorities involved in water resources management and to the effectiveness of the AIS. Arrangements made in the past concerning the use of water (water rights) should be identified, since these may significantly constrain the options for water resources development.

Water resources management studies are often limited to the preparation of policies for a certain

agency. In this situation, the analysis of the AIS will mainly serve to identify measures that the agency can implement effectively. The responsible agency should be aware of the possible role they may have in solving the management problems. Sometimes, the analysis of the AIS may result in recommendations for institutional and legal changes.

13.4.2.2 Data and Modeling

The result of the data collection and modeling activities is a quantitative representation of the WRS at an appropriate level of detail. The framework is designed to assess the effects of individual measures or combinations of measures, expressed in values for the evaluation criteria chosen. If computer programs for running models have to be developed or if existing computer programs have to be adapted in a significant way, a considerable effort may be required which may consume a large part of the available budget and time. Careful selection of the phenomena to be represented by the models, tuned to the needs of the project, is important.

During the modeling activity, more information on the study area and the type of measures to be considered may become available. This could lead to changes in model structure. The models should therefore be flexible and adaptable to new information.

Model Integration

The various models and components developed for the NRS and SES describe parts of the total system. Some models may produce output that is needed as input for another model. For example, the output of a water quantity model may be the input to a water quality model requiring different spatial and temporal resolutions. Some models may include links to various sub-models and run interactively, others not. Depending on the models and the problem situation, single or multiple linked models may be included within an interactive decision support system. In other cases, a clear description of information flow from one independent model to another may be sufficient.

Figure 13.8 provides an example in which various simulation models are combined to

analyze a river basin under drought conditions. The reservoirs in the system involve sedimentation and hydropower generation. The core of this modeling framework is formed by the “core models” block in the upper right corner of the figure. In this block the demand for water is determined, followed by a balancing of supply through water allocation decisions. Links among these core models are automatic. Other models are linked through file transfer. This applies to the required input on macroeconomic and hydrometeorological conditions (generated by scenarios) as well as the side analysis of the sedimentation and water quality in the reservoirs. The last parts of the computational framework are the modules that determine the financial and economic aspects (investments, operation and maintenance, benefit–cost, etc.) and support a multi-criteria analysis.

At various places in this modeling framework, one can change the values of input parameters. Scenarios can be analyzed by changing the macroeconomic and hydrometeorological conditions.

Figure 13.8 is just an example. Other problem situations may require different modeling frameworks. The goal in creating such model frameworks is to make them as simple and transparent as possible, and still adequately address the problems to be solved. Sometimes complexity is necessary. In any event it saves time and money to start as simple as possible and only add more detail when necessary to carry out a proper analysis.

Collaborative modeling

Involving decision-makers and stakeholders in the analysis process has till recently been limited to the more general analysis about problems and solutions. The quantitative information, e.g. resulting from models, was provided by the analysts (e.g., consultants) as input for the discussions. More and more we see that stakeholders do not accept this black box approach anymore. They want to understand what went into the model, how the models work and, preferably, they want to “play” with the model themselves. This is a promising development as this will increase the

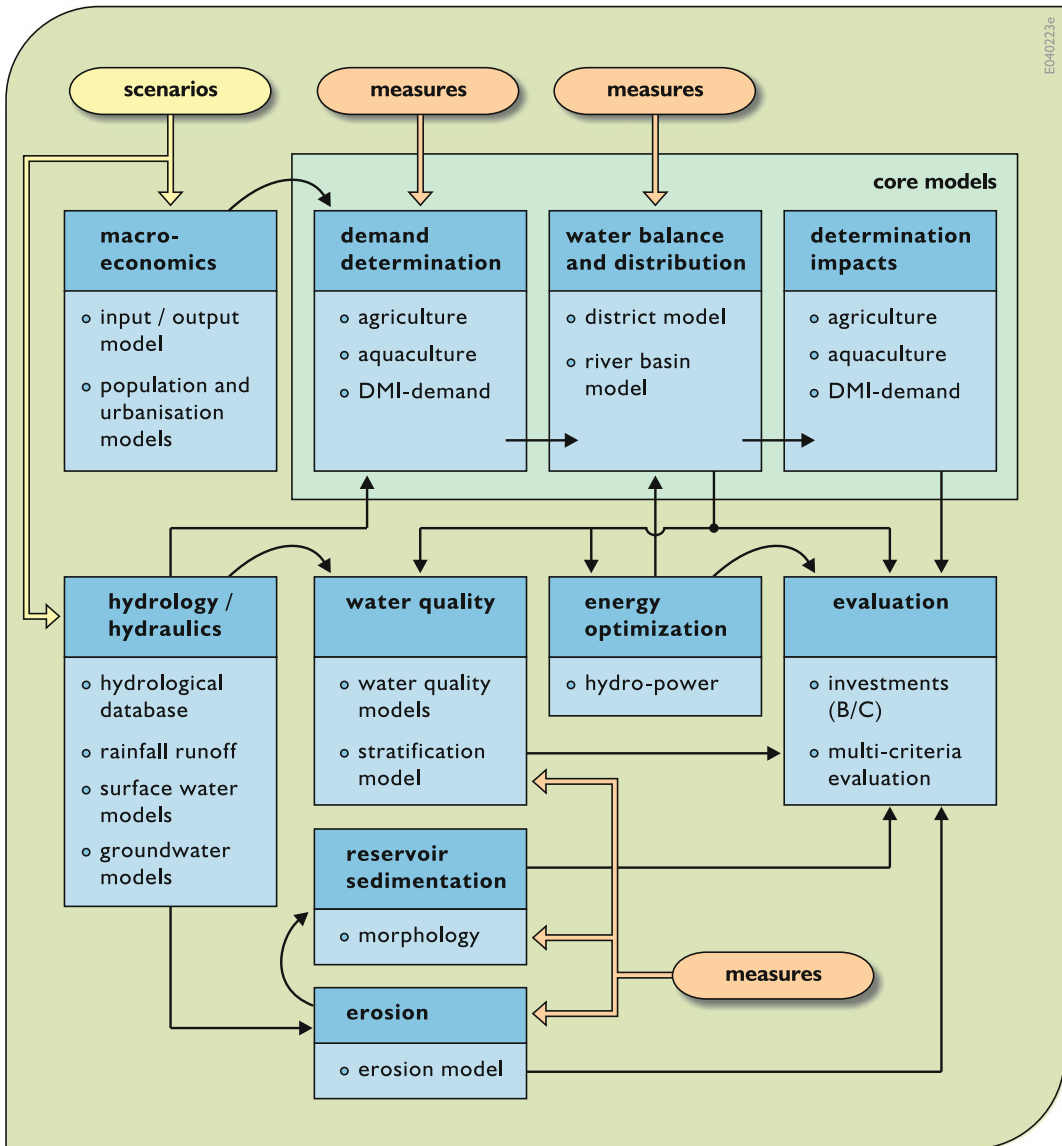


Fig. 13.8 Example of typical computational framework of simulation models

understanding of the stakeholders on how the system works and let them see the opportunities and constraints of that system. Having stakeholders involved in the development and running of the models requires that these models are made more accessible and intuitive, in particular their input/output interfaces. It requires also a different attitude of the modelers. Various approaches to

collaborative modeling are currently being developed, sometimes under different names such as Collaborative Modeling for Decision Support (e.g., shared vision modeling), Mediated Modeling, Group Model Building, Companion Modeling, Interactive Modeling, Networked Environments for Stakeholder Participation or Model-supported Collaborative Planning.

13.4.2.3 The Need for a Structured Quantified Analysis Process

Decision making on measures and strategies to improve the performance of the WRS should be based on quantified information about the present problems (e.g., average flood damage) and the impacts of proposed measures (e.g., the reduction in flood damage) and the costs of these measures. To be able to produce this quantified information the following is needed:

- a structured analysis process (this section); and
- a computational framework (see previous section).

The analysis process starts with a quantified problem description. The analysis of the present situation is called the Base Case analysis. To be able to predict possible future problems scenarios should be defined on how this future might develop. The computational framework will calculate the impacts (the future problems) of these possible external developments. This is often called the Reference Case analysis.

Base case

The performance of the WRS is studied for the infrastructure and water demands in the base case. The base case is based on the base year, which is the most recent year for which a complete set of data can be collected. The base case describes thus the performance of the WRS in the present situation. A comparison of the base case with the criteria (and possible targets) specified in the WRM objectives will result in a quantified problem statement.

Scenario conditions

A good plan should also address the expected water-related problems in the future. The analysis for the future time horizon(s) should include different scenario conditions. Possible scenario conditions for WRM are socioeconomic developments (change in demand and pollution) and climate change (including sea level rise). See the

next section on more information about developing scenarios.

Reference case

The reference case addresses the future situation by considering the present infrastructure, to which measures are added that have already been decided or are being executed, together with selected scenario conditions. In the reference case an analysis of the performance of the WRS is undertaken if present policies and regulations are continued and followed by the government and the water users.

Problem description—present and future

The problem description should be carried out based on the results obtained from the base and reference case analyses in combination with the problems and issues perceived by the decision-makers and stakeholders. A problem analysis should be expressed as far as possible in terms of the socioeconomic and environmental impacts that have a meaning to the decision makers and stakeholders. An integrated approach is crucial for a solid understanding of the system and its associated problems. The integrated approach can only be achieved if the plan defines the main problems and issues in the basin and its interlinkages. For this, it is important that the plan is aligned with other related plans such as Watershed Plans (erosion), Flood Risk Management (FRM), and Integrated Coastal Zone Management (ICZM), amongst others.

Inventory of potential measures and selection of promising measures

Once the present and future problems are known measures (including “no regrets” that can immediately be implemented) can be identified that will address these problems. An inventory should be made of all the measures that the stakeholders are planning or considering. Based on the quantified problem analysis additional measures might be formulated. The computational framework can be used to determine the impacts of these measures. The most promising measures will be kept for detailed analysis in the next step: Strategy Building.

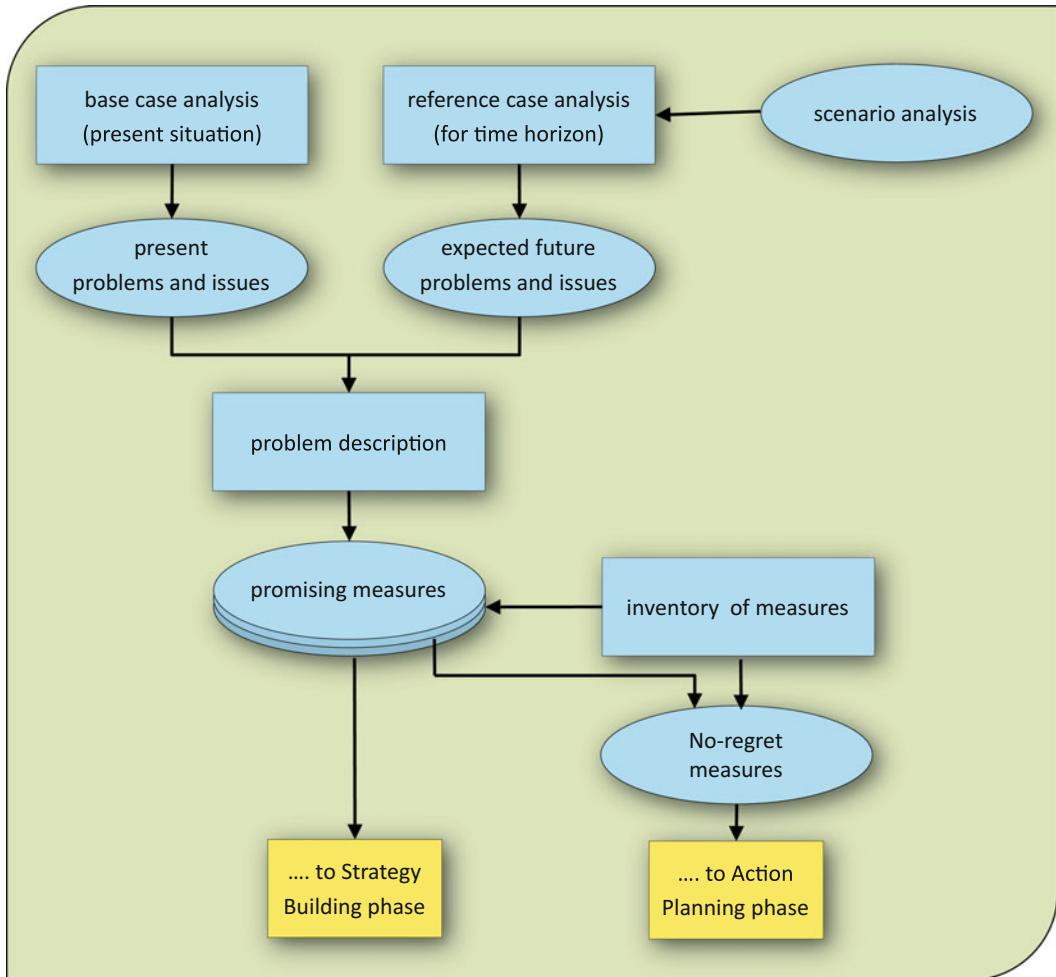


Fig. 13.9 Structured analysis process

The above described structured analysis process is illustrated in Fig. 13.9.

13.4.2.4 Scenario Analysis

A good plan should not only address the present problems but should also prepare for problems that might arise in future. To predict the future scenario assumptions have to be made. Scenarios are possible developments external to the WRS, i.e., outside the control of the decision makers involved in the project. The most usual scenario components for water resources studies are socioeconomic developments (e.g., growth of population and economic activities) and climate change (including sea-level rise). For the

economic evaluation of the plan it might be needed to make assumption about the future prices of energy and food. Changes in diet (e.g., the consumption of more meat) can also be important.

The most used combination of scenario elements are presented in a quadrant of low and high economic growth versus slow and fast climate change. Ideally the whole analysis should be carried out for all kind of scenario combinations and the selection of the best strategy should be based on the evaluation which strategy is able to cope with all these possible future developments. In reality most analyses are carried out for the most likely scenario based on a trend analysis

or Business-As-Usual (BAU). The strategy that follows out of this is then analyzed in a “scenario analysis,” to test that strategy on robustness and flexibility for other possible futures. See also Sect. 13.4.3.2 on adaptive management analysis.

13.4.2.5 Quantified Problem Analysis

A problem analysis should address and be expressed in terms of the socioeconomic and environmental or ecosystem impacts that are of interest to the decision makers. Not all stakeholders may be able to relate to predicted changes in flows, water levels, or pollutant concentrations. Some may want to know how much money is involved, the rate of shore line erosion, the relative change in fish population, or the number of people affected by flooding. Expressing outcomes in terms of socioeconomic impacts makes it easier to relate the problems to the (socioeconomic) development objectives that decision-makers have formulated for the particular region or system under consideration.

A good problem analysis will also indicate the measures that can be taken to eliminate, reduce or alleviate the identified problems or to take advantage of new beneficial opportunities. The identification of measures not only helps to clarify the problems and possible solutions; but also helps in the design of the computational framework and the data collection activities. These activities should be designed in such a way that the measures can be evaluated in the analysis phases of the study.

On completion of the initial analysis, project staff (and the decision makers/stakeholders) should have a clear idea about what will be studied in subsequent phases, for what purpose and under what conditions.

13.4.2.6 Identification and Screening of Potential Measures

Once the base and reference cases have been defined, and the problems and bottlenecks identified, measures to address resource management problems can be considered. Measures can be divided into different categories. An inventory of

all possible kinds of actions that can be taken will in general result in hundreds of discrete possibilities. In most cases it will not be practicable to analyze all of them in detail. A screening process is needed to select the most promising ones. This can be done in several ways. As mentioned in various chapters of this book, separate optimization models can be used to eliminate less attractive or less promising alternatives. It can also be done by using the modeling framework developed for the project but limiting the analysis to a few criteria, such as economic or environmental ones. A third kind of screening analysis is to apply judgment as to criteria effectiveness, efficiency, legitimacy and sustainability. Box 13.3 describes these criteria.

Box 13.3. Criteria for screening

Effectiveness. Measures to be taken are those which solve the most serious problems and have the highest impact on the objectives. Measures to prevent problems will be preferred to those that solve them. Similarly, measures that solve problems will be preferred to those that only control them.

Efficiency. Measures to be taken should not meet the explicit objectives at the expense of other implicit objectives. The cost–benefit analysis (at the national level) is one indicator of efficiency. An example is to create a law that forces industrial firms to incur the full cost of end-of-pipe wastewater treatment. In Egypt, this would improve the Nile system water quality, and thus improve health of those who drink it and reduce environmental damage. On the other hand it might impose high costs to the firms, possibly resulting in loss of employment. An efficient decision may be to opt only for cost sharing rather than full cost recovery.

Legitimacy. Measures to be included in the strategy should not rely on uncertain legal/institutional changes. Measures should also be as fair as possible, thus

reducing public opposition so that they will be favored by as many stakeholders as possible.

Sustainability. Measures to be taken are those that improve (or at least do not degrade) the present environmental and socioeconomic conditions for future generations.

The aim of the screening process is to identify those measures that should be further analyzed. The screening of measures is a cyclic process. Assessing the measures will contribute to a better understanding of their effectiveness and new ones may be identified (comprehension loop). Combinations of measures may be considered for specific parts of the WRS, for instance for solving the water quality problems in a subbasin. The result of the screening process is a set of promising measures that can be used for strategy design. The whole process of base case and reference case analysis and screening is depicted in Fig. 13.9.

No regrets

A special category of promising measures are the “no regrets.” More realistic we should speak of “likely no regrets” and “low-regret” measures. These are measures on which there is a very large agreement among the decision-makers and stakeholders that these should absolutely be implemented, preferably as soon as possible. It should be ascertained that these measures will not have negative impacts on other measures or will prevent other possible promising measures to be implemented. The reason to define such no regret measures is that in quite some situations there is a huge pressure to actual implement measures and not to wait till (another) big integrated study has been completed and accepted in its full extend. In particular in developing countries there is a big need for proposals for such measures. These measures can proceed immediately to step IV on Action Planning.

13.4.3 Step III—Strategy Building

In the Strategy Building step, promising measures are combined into strategies. The effects of various strategies are assessed and a limited set of promising ones is defined. For these promising strategies, the effects are assessed in more detail. The sensitivity of these effects to the values assigned to the uncertain model parameters is then assessed. Finally, the results of the selected strategies should be presented to the decision-makers. The selection process is depicted in Fig. 13.10.

13.4.3.1 Strategy Design and Impact Assessment

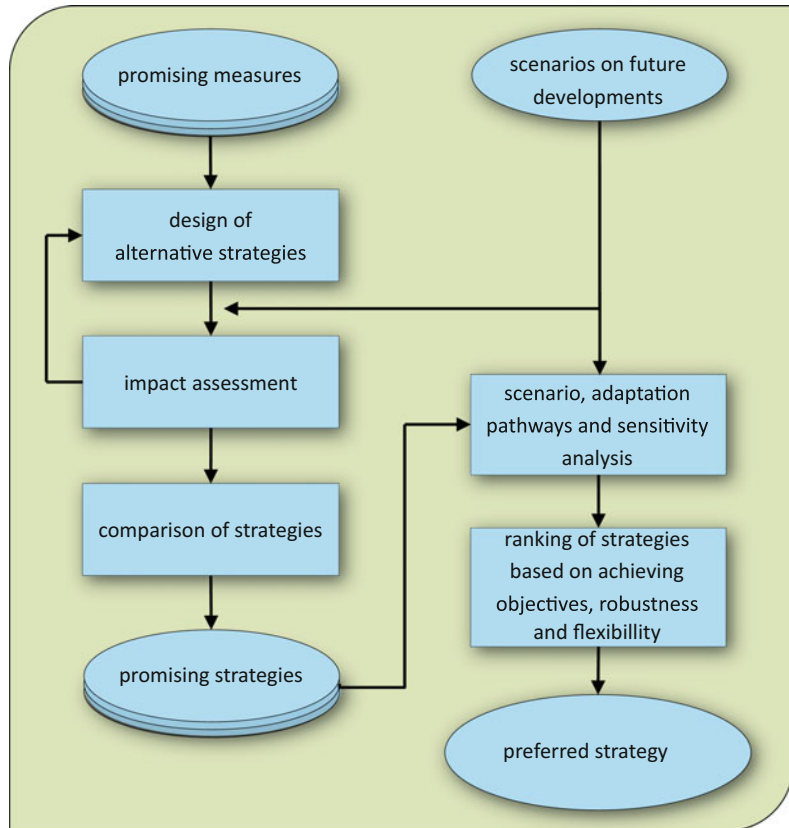
Strategy design involves the development of coherent combinations of promising measures to satisfy the management objectives and meet the management targets if possible. As there are generally many criteria related to these objectives, and probably many expressed in different units, strategy design is not a simple process. Relations among combinations of measures and their scores on the evaluation criteria are complex. The optimum combination may depend on who is asked. Trade-offs among the values of different criteria, and disagreements among various stakeholders, are inevitable.

The design of strategies is an iterative process. One can start by developing strategies on the basis of a single objective such as, for example, reliability of food and energy production or maximum net economic benefits. These strategies define the boundaries of the solution space. Comparison of the impacts of these strategies can lead to the construction of compromise strategies by changing elements in the strategy. A resulting loss with respect to one criterion is then compared with gains to another.

Evaluation of Alternative Strategies

Strategies can be compared based on their criteria values or scores. To facilitate the comparison, the number of evaluation criteria should be limited. Criteria have to be comprehensive (sufficiently

Fig. 13.10 Activities in the strategy building phase



indicative of the degree to which the objective is met) and measurable, i.e., it should be possible to assign a value on a relevant measurement scale. Where possible, criteria should be aggregated; for example, some financial criteria might be processed into a single value when distribution issues are not going to be important.

It is usually impossible to express all criteria in a single measurement scale such as a monetary value. (We say this recognizing the many attempts to do so by highly respected economists.) Criteria related to environmental quality or ecosystem vitality or the beauty of a scenic view can often be expressed quantitatively but in nonmonetary terms. This should, however, be done in such a way that a ranking is possible on the basis of the chosen criteria.

Generally, there will not be a single strategy that is superior to all other ones with respect to all criteria used in the assessment. That means that an evaluation method is required for the ranking of alternative strategies.

Scenario and Sensitivity Analysis

Before drawing conclusions from planning projects involving uncertain information, and indeed predictions of possible futures, one should analyze the effects of changes in the uncertain assumptions made throughout the analyses. If the selection of a different scenario would significantly change the attractiveness of a selected strategy, then additional study may be required to reduce the uncertainties in that scenario. The sensitivity of the results to changes in model

parameter values and assumptions should be determined and addressed in a similar way.

13.4.3.2 Adaptive Management Analysis

The analysis approach described in the previous section is based on the assumption that it is known what will happen in future. Predictions are made on how population growth, economic growth, spatial developments (e.g., urbanization) and climate change will take place. Some of these developments are quite certain, e.g., population growth for which one can make reasonable good projections. Other developments are much more uncertain such as economic growth and climate change. While we want to be prepared for these future conditions we do not want to run the risk that huge infrastructural investments are being made which later appear to have been overdesigned or even unnecessary.

The way to deal with future uncertainty is to follow an adaptive management approach. An adaptive management approach has to replace the traditional approach of master plans for the

basin. The development of implementing stand-alone projects to adaptive management is illustrated in Fig. 13.11.

The message on how to follow an adaptive management approach is given in the right two columns of Fig. 13.11 and is the logical follow-up from the project oriented developments in the two first columns. The figure explains that:

- The project-based approach is straightforward and easy to implement. This approach does not consider the (positive and negative) interaction of the project with other projects.
- The interaction is taken into account when related projects are considered in a package of projects. However, the overall system is not integrated yet and not optimized.
- The traditional master planning tries to optimize the overall system. The projects are implemented as components of an integrated strategy. The implementation of the strategy includes an optimization of the various projects over the planning period which is usually between 15 and 30 years, for which a cost-benefit analysis usually applies. Such a

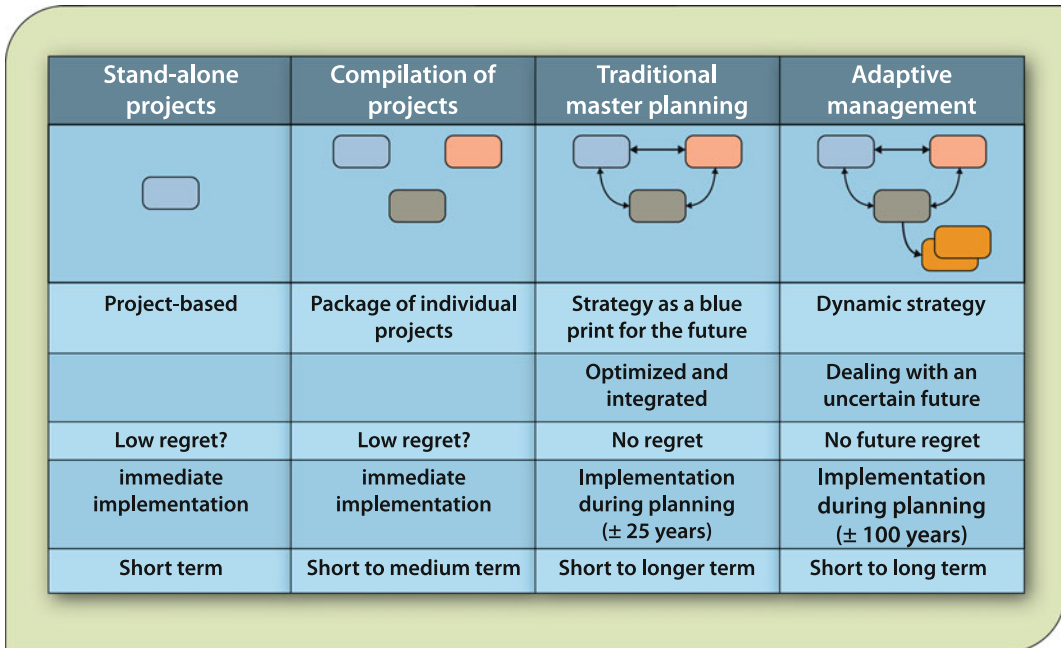


Fig. 13.11 Planning approaches in water resources management

- master planning approach does not consider the long-term uncertainties that are involved in socioeconomic developments and climate change. If the predicted changes in socioeconomic conditions and climate do not materialize this might lead to “future regret.”
- To reduce future regret a planning period of up to 50 or even 100 years needs to be considered. As the lifetime of most structural measures (dikes, floodways, reservoirs, etc.) are designed for a period of 50–100 years, it is wise to incorporate future uncertainties in boundary conditions in their designs and make them part of a dynamic strategy. The adaptive approach not only tells us what to do now but also gives directions on what to do when the conditions develop differently.

Adaptive pathways

Various methods have been developed that enable us to deal with future uncertainties. Recent methods include Decision Trees (Ray and Brown 2015) and Dynamic Adaptation Policy Pathways (DAPP; Haasnoot et al. 2013). The Decision Trees is a repeatable method for evaluation of

climate change risks to new development projects. DAPP identifies tipping points that determine in time when a certain policy or action is no longer acceptable and (another) action is needed. By exploring all possible actions you can develop adaptation pathways that will minimize the regret. The Adaptive Pathway Approach is illustrated in Fig. 13.12. The approach requires that many conditions are explored (pathways, scenarios, long time series). For that reason the models used in an adaptive pathway analysis are sometimes limited versions (meta-models) of the ones described in this book. See Haasnoot et al. (2014). Following an adaptive pathways approach basically means that two additional criteria should be considered in decision-making:

- **Robustness:** how robust is the existing strategy when the future develops differently than expected? Will the strategy then still achieve the objectives?
- **Flexibility:** how changeable is the strategy when it appears that the future develops differently than expected and we need to change the strategy?

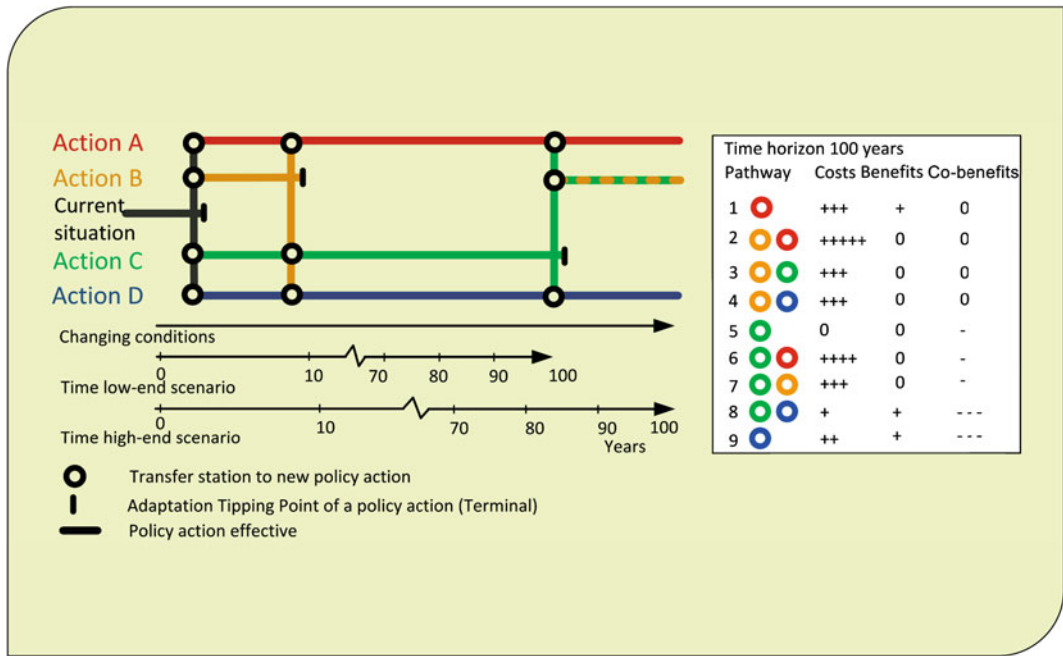


Fig. 13.12 Adaptive pathways approach

Robustness and flexibility often have a strong relationship with costs. A robust strategy can be more costly (big reservoirs, high dikes, etc.). A flexible strategy (many small reservoirs, build in time) can also appear to be more expensive in the end. These costs need to be taken into account when deciding on a strategy.

13.4.3.3 Presentation of Results—Preferred Strategy

Presentation of the selected promising strategies to decision-makers may be by means of briefings, presentations, and summary reports among other means. The level of detail and the way project results are presented should give an overview of the results at an appropriate level of detail for the audience involved. Visual aids such as score cards and interactive computer presentations of study results are often very helpful for promoting a discussion of the results of the analysis.

The results of selected strategies can be presented in matrix form on “scorecards.” The columns of the scorecard represent the alternative cases used in the analysis. The rows represent the impact of different alternatives with respect to a given criterion. An example is depicted in Table 13.2. Scorecards can contain numbers only, or the relative value of the criteria can be expressed by plusses and minuses, or a color or shading. The purpose of scorecard presentations is to present a visual picture of the relative attractiveness of the alternatives based on various criteria. Scorecards can also help viewers detect clusters of criteria for which alternatives have a consistently better score. The presentation of the results in scorecards allows a decision-maker to give each impact the weight he considers most appropriate.

13.4.4 Steps IV and V—Action Planning and Implementation

Once the preferred strategy has been selected this strategy should be translated into concrete actions. Careful planning and coordination is

required as many authorities will be involved in the implementation. The action plan will have an “open” and “rolling” character, meaning that it is not static or prescriptive, and leaves room for individual decision-makers to further elaborate upon in relation to their own responsibilities. On the other hand, the action plan should be concrete, by assigning clear responsibilities for carrying out the activities involved. It also should include the budgetary requirements for the implementation, including investments and recurrent costs.

13.4.4.1 Investment and Action Plan

The action plan translates the selected strategy in concrete actions. For each of these actions it should be clear:

- what: concrete actions that have to be carried out for each of the measures included in the strategy to get it implemented?
- who: the prime decision-maker/stakeholder responsible for carrying out the action and who will take the lead in the implementation;
- how: the steps to be taken and the consultative process involved;
- when: the time planning; and
- financing: where will the money to implement the action come from?

What

An integrated planning analysis is usually carried out at pre-feasibility level. A rough description of the measures will be included in the strategy and the assessment was based on first estimates of costs and benefits. Depending on the type of measure, feasibility studies should be completed before the measures can actually be implemented. Often these feasibility studies are combined with detailed (technical) design of the measures.

Who and How

The Action Plan aims to stimulate the coordinated development and management of the water resources. This is illustrated in Fig. 13.13, which presents the Implementation Plan for water resource development in Central Cebu in the

Recommended Measures			Phase	Institutions Involved																	Implementation Info							
				Province of Cebu	City of Cebu	MCWD	DBO, COA, MRE, LULA, ...	City of Marikina	City of Lapu-Lapu	City of Iloilo	City of Danao	Municipality of Naga	Municipality of Marikina	Municipality of Lucena	Municipality of Consolacion	Municipality of Bataan	Municipality of Pangasinan	Municipality of Alcala	Municipality of Carmen	Municipality of Compostela	Private investment	NGOs, academe, church, etc.	No. of the measure	Type of measure	Phase I only			
																							Total investment (in MPhp)		Recurrent costs (MPhp/yr)	Starting year effective operation	Needed leadtime (preparation + construction)	
Develop more resources	West	Groundwater wells West	I			X																9	invest	849	38	2006	-	
	Uplands	Spring boxes + small impoundments + trucking for drought periods	I			X																8	invest	737	126	2006	-	
	East, non-MCWD area	Groundwater wells East	I			X																9	invest	639	28	2006	-	
	East-MCWD area, excl. Mactan	Groundwater production wells by MCWD	I			X																9	invest			2006	-	
		Groundwater wells MCWD area by private initiatives	I			X																9	invest	588	26	2006	-	
		Luyang-Carmen weir	I		X		X							X	X					X	X	4	invest	3 169	124	2016	2+2	
		Northern well fields: Liloan, Compostella, Kadiot, Danao, Carmen, Luyang (excl. Luyang weir)	I			X																9	invest			2006	2+2	
		Lusuran Dam	I			X																2	invest	2 536	77	2027	6+2	
		Kokot Dam	I			X																3	invest	905	59	2022	6+2	
		Southern well fields: Napo/Cancar river, Pangdan, Merganilla	I			X		X														10	invest	2 490	86	2014	2+2	
	Mactan	Horizontal wells	I			X		X														11	invest	51	5	2006	-	
		Shallow fresh water wells	I			X		X														11	invest	276	56	2008	-	
		Shallow brackish water wells + reverse osmosis	I			X		X														13	invest	80	42	2006	-	
		Desalination by MCWD for industry	I			X		X														13	invest	297	55	2010	2+2	
		Desalination by MCWD for domestic use	I			X		X														13	invest	1 393	167	2012	2+2	
		Desalination by industry (private initiative)	I			X		X														13	invest	297	55	2010	2+2	
Water demand reduction	Reducing losses	Rainfall harvesting: urban, rural, industry	I																		12	invest	515	-	2006	-		
		Leakage reduction / rehabilitation distribution system MCWD	I																		14	invest	820	41	2007	-		
		Adequate and differentiated water pricing	III			X															20	study	pm	pm	2007	1		
	Awareness	Promotion of water saving equipment and less consuming production	II																		21	man	-	-	-	-		
		Awareness raising high demand Cebu neighbourhoods	I																		22	man	0	2.6	2006	-		
Protecting the resources	Watershed management	Awareness raising - general	I																		23	man	0	6.9	2006	-		
		Land use practices / watershed management	III																		30	reg	-	-	-	-		
	Water quality improvement	Gabion dams	III																		31	invest	-	-	-	-		
		Improved solid waste management	II																		34	man	-	-	-	-		
		Well head protection - spatial planning recharge areas	II																		35	reg	-	-	-	-		
		Prevent sand and gravel mining in rivers (enforcement)	II																		36	reg	-	-	-	-		
		Implement strict effluent permitting (EME)	II			X															37	reg	-	-	-	-		
		Urban sewage systems in building requirements	I																		38	reg	0	0	2007	1		
		Development of urban sewage systems and treatment	II																		39	invest	-	-	2010	3		
		Sanitary programs in uplands	III																		40	invest	-	-	-	-		
Water governance	Organisational	Development institutional setting for MWRM (integration and coordination - Board, TS etc.)	I																	50	instt	pm	pm	2006	0			
		Family planning and migration control (env. related)	I																	54	man	pm	pm	-	-			
	Regulation and control	Water withdrawal quantity and quality control	II																	52	man	pm	pm	2007	0			
		Implement and enforce spatial planning (urban, industrial, etc.)	I																	55	man	pm	pm	-	-			
Research and development	Research and development	Priority rules (allocation) during dry periods	I																	56	reg	pm	pm	-	-			
		Improve data collection, analysis and presentation/improve decision support tools	II																	61	capac	pm	pm	2007	-			

Fig. 13.13 Implementation plan (taken from Cebu study)

Philippines. The measures included in the plan will involve or affect many stakeholders. All these stakeholders (based on the outcomes of the stakeholder analysis and designed participatory planning process) should therefore be included in some way in the implementation process in order to guarantee a successful implementation and a

sustainable benefit of the particular measure. In general the following roles can be distinguished:

- **Responsible:** the stakeholder has the first responsibility for the implementation of the measure but will co-operate with and/or consult other stakeholders in this process. In

Fig. 13.13 this is indicated by the symbol: “●”.

- Co-operate: the stakeholder has an important say in the implementation of the measure but is not the first responsible and is expected to work with other stakeholders in this matter. In the figure this is indicated by the symbol: “○”.
- Consult: the stakeholder has an interest in the implementation of the measure and will be consulted by the first responsible. In certain cases permission will be needed before the implementation can take place. In the figure this is indicated by the symbol: “x”.

When

The action plan should also specify the timing of the implementation. When will (the preparation of) the implementation start, and when should the implementation be finalized. This information is needed for the overall investment plan but also because some measures will depend on the completion of other measures.

13.4.4.2 Financing—Investment Plan

An important, if not the most important, part of the Action Plan is to determine how the action will be financed. The sources of the financing will largely depend on the type and size of the measure. As water resources management is mainly a governmental task, most of the finances will come from public sources. These can be from the national budget (possibly supported by donor funds) or

from local (province, municipality) budgets. In some cases private funding can be considered in PPP (Public Private Participation) constructions. This seems in particular attractive when there is a good possibility for payment by the stakeholders of the services that will be provided. Examples where PPPs can be considered are urban public water supply and hydropower production.

The investment plan should also address how the recurrent costs (operation and maintenance) of the implemented projects will be recovered. Preferably this should be done based on fees to be paid by the people that benefit from the project.

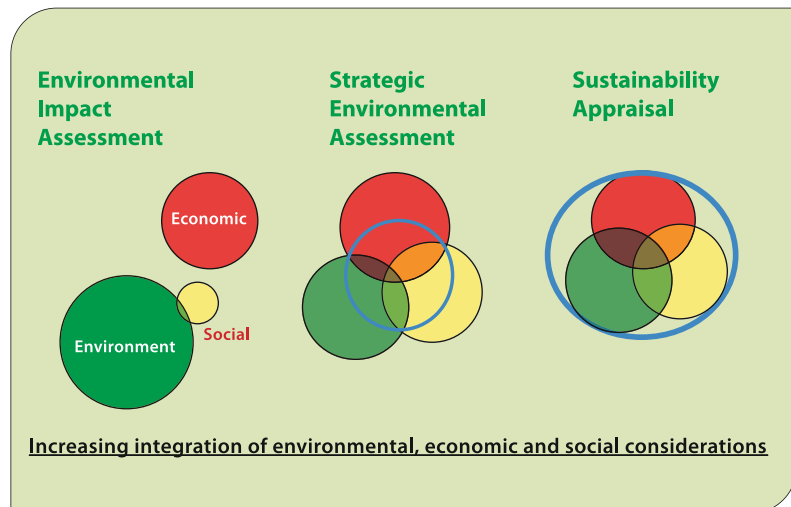
13.4.4.3 Feasibility Studies and Environmental Impact Assessment

A feasibility study should include a more detailed study of the projects (measures) proposed in the plan. Commonly a feasibility study includes some 5 areas of feasibility:

- technical
- social/environmental
- political/legal
- financial/ economic
- operational and scheduling

A feasibility study for a good implementation planning will often include a more detailed assessment of the possible socioeconomic and environmental impacts of some of the measures

Fig. 13.14 Applying SEA (OECD 2006)



that comprise the preferred strategy. There are several types of assessment depending on the focus of the study. As depicted in Fig. 13.14 the most well-known are: Environmental Impact Assessment (EIA, for infrastructure projects), Strategic Environmental Assessment (SEA, mainly used in policy development) and Sustainability Appraisal (SA).

13.4.4.4 Promotion

After the action plan has been established one needs to find ways to increase the influence of stakeholder groups that favor the implementation of the action but lack influence; to change the attitude of influential groups that are opposing this action; and to use the positive attitude of influential groups that are in favor of this action. The results of the stakeholder analysis are used for the identification of the stakeholder groups. As illustrated, the matrix highlights the strategy toward project acceptability or appreciation and therefore smooth implementation.

To create maximum awareness, enthusiasm and support for selected projects within the Action Plan the selected stakeholder groups need to be provided with the right information on the project. Additionally, involving a selection of stakeholders in project preparation and

implementation will assist in making them enthusiastic about the project. To do this effectively, a mix of marketing options can be used. Appropriate marketing options might be:

- mass one-way communication for the general public (such as newspapers, radio, television plus more traditional media in the more rural areas);
- selective one-way communication for selected stakeholders groups (direct mail, brochures with more specific information dedicated for the selected group); and
- personal two-way communication between the project promoter and selected stakeholders groups (education method, outreach method or more risky word-of-mouth method).

13.4.4.5 Monitoring and Evaluation

An overview of the implementation framework is given in Fig. 13.15. This implementation framework applies for both Steps IV (Action Planning) and V (Implementation). The actual implementation of most of the measures will take place by decentralized agencies of national ministries or at local governmental level and their related utilities, districts, and associations. Where

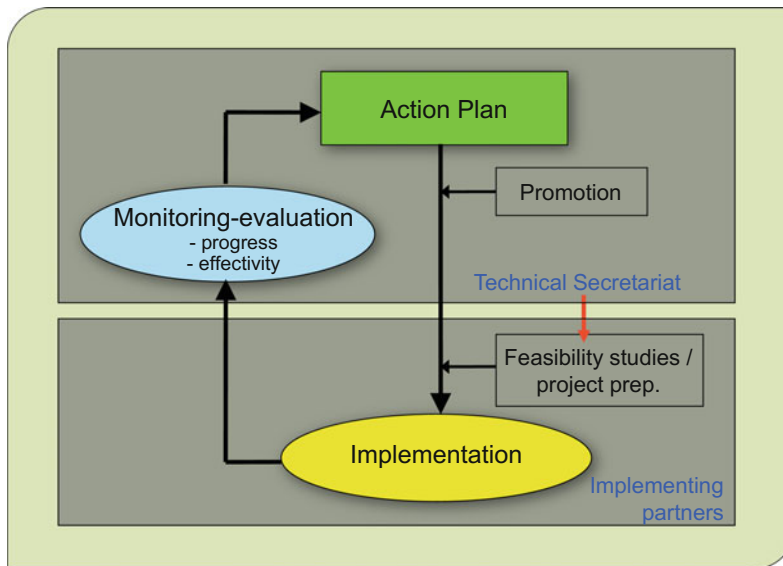


Fig. 13.15 Implementation framework

needed feasibility and engineering studies will be carried out before the actual implementation and/or construction can take place.

Above the implementation level there should be a guidance and coordination level, e.g., a Technical Secretariat (TS) at basin level. Periodically a monitoring report compiled by the TS can track the progress made in implementing the measures of the Action Plan and the effectiveness of these measures in meeting their objectives. Insufficient progress may lead to an adjustment of the Action Plan. The TS may also provide assistance to the implementing partners, e.g. the local government agencies, as they carry out feasibility studies. The TS should be able to support them by providing data and possibly other relevant information from their Management Information System (MIS).

13.5 Making It Work

The framework of analysis presented in Fig. 13.3 includes next to the five steps of analysis two crucial blocks that play a role in several of these steps and deserve special attention. The first one is the stakeholder engagement in the analysis. Involving stakeholders and making sure that their ideas and suggestions are taken into account is an absolute requirement to develop a consensus and support for the ultimate plan that is to be implemented. There is no guarantee that a consensus will be reached, however. Involving stakeholders in each stage of the planning framework takes extra time and money, but if any ultimate plan is to be accepted and prove sustainable, there is no other choice. At a minimum, any plan that is derived from this process should be an informed one, based on inputs from all affected stakeholders and decision makers.

13.5.1 Stakeholder Engagement

The stakeholders that should be involved in a planning process will depend on the specific basin

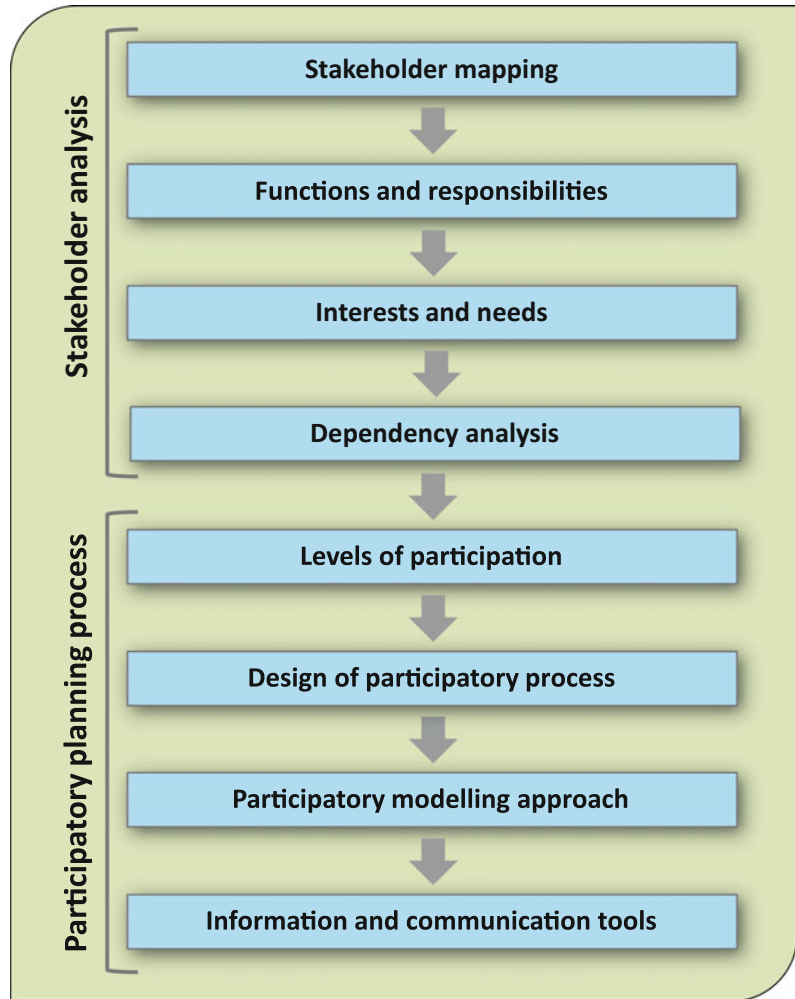
that is being addressed. In general the stakeholders will be all people and/or organizations that:

- will be effected by the plan; and
- are needed to implement the plan.

An integrated plan and its implementation depend to a large extent on the acceptance and ownership of the plan by the decision makers and stakeholders at national and basin levels. A participatory planning process is therefore indispensable for sustainable WRM. A participatory planning process is the results of a set of steps, as depicted in Fig. 13.16. However, the order of the steps can vary according to the local situation and conditions. The prerequisite for the design of a participatory planning process is a good stakeholder analysis. The stakeholder analysis is a supporting planning tool that supports the identification of stakeholders and its engagement. Particularly, this analysis technique supports the task of identifying and in some occasions classifying the stakeholders according to their functions, capacities, interests, concerns and needs, as well as their dependencies (including power relations among them).

Based on the results of the stakeholder analysis the participatory planning process is defined. First, it is crucial to define the levels of participation of the various stakeholders. The level of participation of each group of stakeholders varies depending on the stakeholder analysis and on the maximum level of participation that the client of the study wants to achieve. The second step is the design of the participatory process. This will be adapted to the agreed levels of participation and stakeholders involved. The design of the participatory process needs to take into account the modeling approach (informed decision making) so it is carried out in a participatory manner (step 3). Finally, as illustrated in Fig. 13.16, the design of the participatory planning process needs to consider the information and communication tools used for disseminating and communicating the information to the various groups of stakeholders as illustrated in the power-interest matrix of Fig. 13.17.

Fig. 13.16 Steps in a stakeholder analysis and participatory planning process



Stakeholder analysis

A stakeholder analysis provides a better understanding of the perceptions, concerns, roles, interests, and needs of the stakeholders and contributes to a better approach to the solution. It also helps reduce the possibility of forgetting important risks. Finally, this technique increases the chance that the various groups of stakeholders are willing to cooperate in solving the identified problems and issues.

A good stakeholder analysis should contain at least the following steps:

- (1) Situation analysis as point of departure.
- (2) Inventory of the stakeholders involved (e.g., primary, secondary and tertiary stakeholders).
- (3) Mapping of formal relations according to their functions and responsibilities.
- (4) Inventory of interests, perceptions, and needs.
- (5) Mapping of interdependencies.

Levels of Participation

The various stakeholders are grouped into the different levels of participation according to the

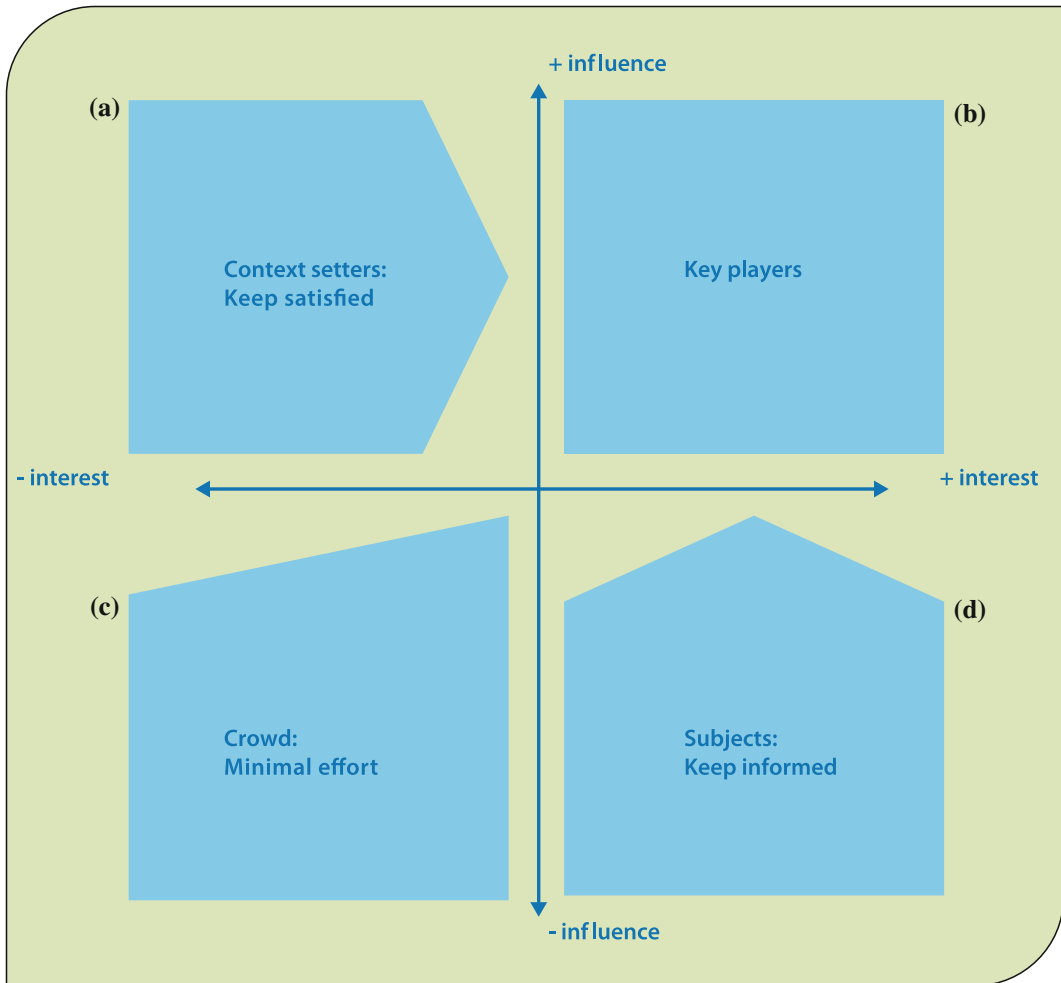


Fig. 13.17 Power-interest matrix

outcomes of the stakeholder analysis, as illustrated in Fig. 13.18:

- Ignorance: where a stakeholder is not aware of what is happening;
- Awareness: where a stakeholder is aware that something is happening;
- Informed: where a stakeholder has been specifically provided with information and is left to decide what to do with it. The emphasis is on the one-way provision of information, with no formal option for the stakeholder to provide feedback, negotiate, or participate in the decision-making process;
- Consultation: where a stakeholder is asked to provide information inputs to the planning process. Information flows are likewise one-way, but in the opposite direction. That is, information is extracted from stakeholders although no commitment is given to use it;
- Discussion: at this level are fully participating and are asked to give advice and recommendations. Here information flows in both directions between stakeholders operating with different interests and levels of influence, and also between these stakeholders and the organizing team (technical team). Since two-way interactions occur, there is room for

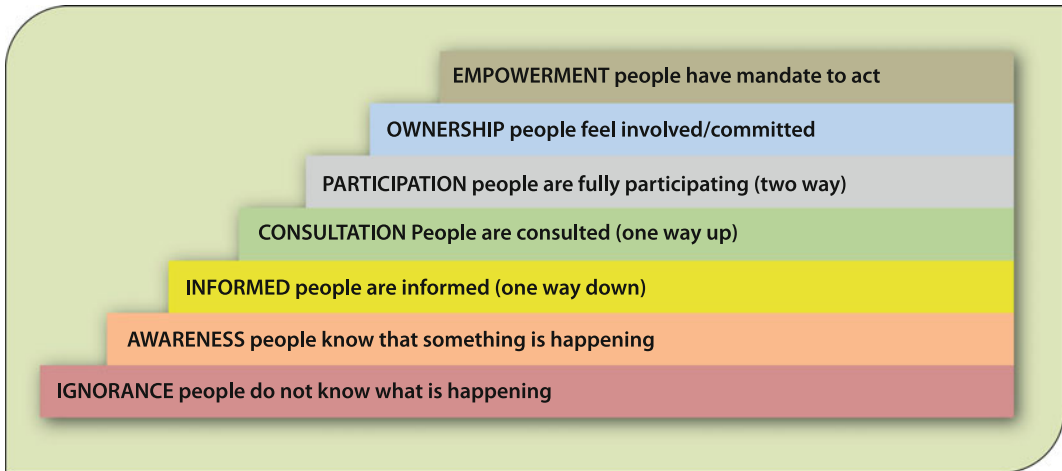


Fig. 13.18 Levels of participation

alternative ideas, solutions and/or strategies to emerge;

- **Co-Design:** at this level stakeholders are actively involved in problem analysis and problem design, which fosters ownership, but where final decision-making powers reside with the governing agencies;
- **Co-Decision-Making:** here decision making powers are shared with those participating stakeholders, leading to their empowerment with respect to the policy/planning decision taken. Typically decisions in these contexts would emerge from a process of stakeholder negotiation.

The first levels (from Ignorance to Consultation) could be thought of as top-down management/planning approaches toward participation, where stakeholders have little control over the decision-making process. The final three levels are more appropriately considered as bottom-up approaches toward participation where stakeholders are much more active and have much more control over the decision-making process.

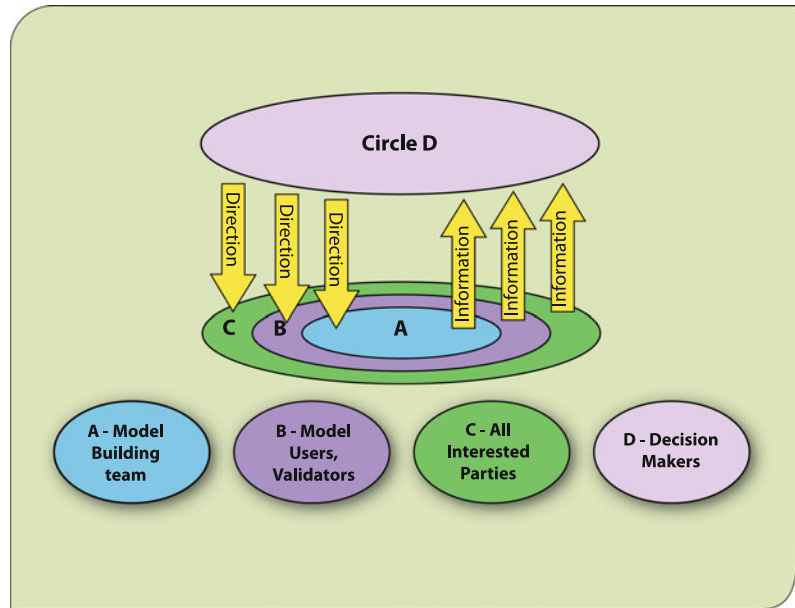
Design of the participatory planning process

The design of the participatory planning process needs to take into consideration the River Basin planning framework and the data and modeling

tools used. Participatory planning tools and techniques enable participants (stakeholders) to influence development initiatives and decisions affecting them. The tools promote sharing of knowledge, building up commitment to the process and empower the group to develop sustainable strategies.

The participatory and informed planning process makes use of the “Circles of Influence” model (Fig. 13.19) that enables to structure participation to limit numbers but not the influence of specific groups of stakeholders (Cardwell et al. 2008; Bourget 2011). Under this model trust is developed in concentric circles; planners and managers work to develop trust with leaders and organizations that other stakeholders already trust. That is, those most directly involved in policy analysis activities (i.e., planners, managers, and modelers who do most of the actual work; Circle A) who communicate with trusted leaders and major stakeholder representatives at the next level (Circle B). These stakeholders then in turn provide a trusted link to all other interested parties, who have much less direct involvement (Circle C). Ideally, Circle B participants would be active in professional or issues-oriented organizations and provide links to others whose interests they represent. Hence, Circle C stakeholders should see their interests represented in Circle B, and have formal

Fig. 13.19 Participatory planning structure based on circles of influence (*source* Cardwell et al. 2008)



opportunities to shape the work of Circles A and B via these representatives. The levels of involvement of those stakeholders in Circle C can vary from Consultation to Awareness. A fourth circle (Circle D) includes decision makers such as agency heads and elected officials, who have been given the authority to accept or reject the recommendations of the policy analysis. For a good participatory and informed planning process it should be clearly identified and engaged throughout the planning process with direction and information flows possible to and from all circles.

Other aspects to be considered for the design of the participatory planning process are:

- Timing of stakeholder involvement. This will be dependent on the Circles of Influence and levels of participation.
- Stakeholder participation in the modeling process (Participatory Modeling). Mainly those stakeholders in the Circles A and B will be regularly involved in some of the phases of the modeling process. The involvement can be concentrated in (i) early and later stages of the modeling process, (ii) construction of the model, (iii) some of the activities prior to model construction, or (iv) only after the final model has been built.
- Type of stakeholder involvement. This can be either individually, with homogeneous (stakeholders with similar interests and problem perceptions) or heterogeneous groups.
- Information and communication tools. Information dissemination (e.g. face-to-face workshops or online platforms) and communication tools need to be adapted to the background conditions of the various groups of stakeholders. This is particularly important for participatory model construction and use, as well as, for the promotion of the plan. The selected marketing options for creating awareness, enthusiasm and support for selected projects within the action plan by stakeholders will vary depending on the results of the stakeholder analysis (Fig. 13.17) and levels of stakeholder involvement (Fig. 13.18). For more information about plan promotion see Sect. 13.4.4.3.

13.5.2 Using Models in a Planning Process

13.5.2.1 Managing Modeling Projects

There are some steps that, if followed in modeling projects, can help reduce potential problems and lead to more effective outcomes. These steps are illustrated in Fig. 13.20. Some of the steps illustrated in Fig. 13.20 may not be relevant in particular modeling projects and if so, these parts of the process can be skipped. Each of these modeling project steps is discussed in the next several sections.

Creating a Model Journal

One common problem of modeling projects once they are underway occurs when one wishes to go back over a series of simulation results to see

what was changed, why a particular simulation was made or what was learned. It is also commonly difficult if not impossible for third parties to continue from the point at which any previous project terminated. These problems are caused by a lack of information on how the study was carried out. What was the pattern of thought that took place? Which actions and activities were carried out? Who carried out what work and why? What choices were made? How reliable are the end results? These questions should be answerable if a model journal is kept. Just like computer-programming documentation, project documentation is often neglected under the pressure of time and perhaps because it is not as interesting as running the models themselves.

Initiating the Modeling Project

Project initiation involves defining the problem to be modeled and the objectives that are to be accomplished. There can be major differences in perceptions between those who need information and those who are going to provide it. The problem “as stated” is often not the problem “as understood” by either the client or the modeler. In addition, problem perceptions and modeling objectives can change over the duration of a modeling project.

The appropriate spatial and time scales also need to be identified. The essential natural system processes must be identified and described. One should ask and answer the question of whether or not a particular modeling approach, or even modeling in general, is the best way to obtain the needed information. What are the alternatives to modeling or a particular modeling approach?

The objective of any modeling project should be clearly understood with respect to the domain and the problem area, the reason for using a particular model, the questions to be answered by the model, and the scenarios to be modeled. Throughout the project these objective components should be checked to see if any have changed and if they are being met.

The use of a model nearly always takes place within a broader context. The model itself can also be part of a larger whole, such as a network

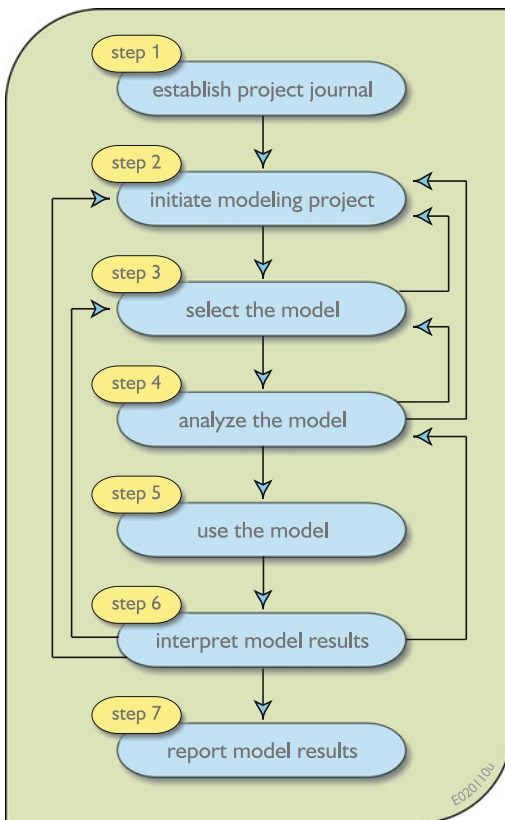


Fig. 13.20 The modeling project process is typically an iterative procedure involving specific steps or tasks

of models in which many are using the outputs of other models. These conditions may impose constraints on the modeling project.

Proposed modeling activities may have to be justified and agreements made where applicable. Any client at any time may wish for some justification of the modeling project activities. Agreement should be reached on how this justification will take place. Are intermediate reports required, have conditions been defined that will indicate an official completion of the modeling project, is verification by third parties required, and so on? It is particularly important to record beforehand the events or times when the client must approve the simulation results. Finally, it is also sensible to reach agreements with respect to quality requirements and how they are determined or defined, as well as the format, scope and contents of modeling project outputs (data files) and reports.

Selecting the Model

The selection of an existing model to be used in any project, as opposed to developing a new one, depends in part on the processes that will be modeled (perhaps as defined by the conceptual model), the data available and the data required by the model. The available data should include system observations for comparison of the model results. They should also include estimates of the degree of uncertainty associated with each of the model parameters. At a minimum this might only be estimates of the ranges of all uncertain parameter values. At best it could include statistical distributions of them. In this step of the process it is sufficient to know what data are available, their quality and completeness, and what to do about missing or outlier data.

Determining the boundaries of the model is an essential consideration in model selection and use. These boundaries define what is to be included in a model and what is not. Any model selected will contain a number of assumptions. These assumptions should be identified and justified, and later tested.

Project-based matters such as the computers to be used, the available time and expertise, the modeler's personal preferences, and the client's

wishes or requirements may also influence model choice. An important practical criterion is whether there is an accessible manual for operating the model program and if help is available to address any possible problems.

The decision to use a model, and which model to use, is an important part of water resources plan formulation. Even though there are no clear rules on how to select the right model to use, a few simple guidelines can be stated:

- Use the simplest method that will yield adequate accuracy and provide the answer to your questions.
- Select a model that fits the problem rather than trying to fit the problem to a model.
- Question whether increased accuracy is worth the increased effort and increased cost of data collection.
- Consider model and computational cost. Today computing costs are rarely an issue except perhaps for some groundwater management problems.
- Do not forget the assumptions underlying the model used and do not read more significance into the simulation results than is actually there.

Analyzing the Model

Once a modeling approach or a particular model has been selected, its strengths and limitations should be assessed. The first step is to set up a plan for testing and evaluating the model. These tests can include mass (and energy) balance checks and parameter sensitivity analyses (see Chap. 8). The model can be run under extreme input data conditions to see if the results are as expected.

Once a model is tested satisfactorily, it can be calibrated. Calibration focuses on the comparison between model results and field observations. An important principle is: the smaller the deviation between the calculated model results and the field observations, the better the model. This is indeed the case to a certain extent, as the deviations in a perfect model are only due to measurement errors. In practice, however, a good fit is by no means a guarantee of a good model.

The deviations between the model results and the field observations can be due to a number of factors. These include possible software errors, inappropriate modeling assumptions such as the (conscious) simplification of complex structures, neglect of certain processes, errors in the mathematical description or in the numerical method applied, inappropriate parameter values, errors in input data and boundary conditions, and measurement errors in the field observations. To determine whether or not a calibrated model is “good,” it should be validated or verified. Calibrated models should be able to reproduce field observations not used in calibration. Validation can be carried out for calibrated models as long as an independent data set has been kept aside for this purpose. If all available data are used in the calibration process in order to arrive at the best possible results, validation will not be possible. The decision to leave out validation is often a justifiable one especially when data are limited. Philosophically, it is impossible to know if a model of a complex system is sufficiently “correct”. There is no way to prove it. [“All models are wrong but some are useful” Box (1976).]

Experimenting with a model, by carrying out multiple validation tests, can increase one’s confidence in that model. After a sufficient number of successful tests, one might be willing to state that the model is “good enough”, based on the modeling project requirements. The model can then be regarded as having been validated, at least for the ranges of input data and field observations used in the validation.

If model predictions are to be made for situations or conditions for which the model has been validated, one may have a degree of confidence in the reliability of those predictions. Yet one cannot be certain. Much less confidence can be placed on model predictions for conditions outside the range for which the model was validated. While a model should not be used for extrapolations as commonly applied in predictions and in scenario analyses, this is often exactly the reason for the modeling project. What is likely to happen given events we have not yet

experienced? A model’s answer to this question should also include the uncertainties attached to these predictions.

Using the Model

Once the model has been judged ‘good enough’, it may be used to obtain the information desired. One should develop a plan on how the model is to be used, identifying the input to be used, the time period(s) to be simulated, and the quality of the results to be expected. Again, close communication between the client and the modeler is essential, both in setting up this plan and throughout its implementation, to avoid any unnecessary misunderstandings about what information is wanted and the assumptions on which that information is to be based.

Before the end of this model use step, one should determine whether all the necessary model runs have been performed and whether they have been performed well. Questions to ask include:

- Did the model fulfill its purpose?
- Are the results valid?
- Are the quality requirements met?
- Was the discretization of space and time chosen well?
- Was the choice of the model restrictions correct?
- Were the correct model and/or model program chosen?
- Was the numerical approach appropriate?
- Was the implementation performed correctly?
- Are the sensitive parameters (and other factors) clearly identified?
- Was an uncertainty analysis performed?

Some of these questions may not apply, but if any of the answers to these questions is no, then the situation should be corrected. If it cannot be corrected, then there should be a good reason for this.

Interpreting Model Results

Interpreting the information resulting from simulation models is a crucial step in a modeling

project, especially in situations in which the client may only be interested in those results and not the way they were obtained. The model results can be compared to those of other similar studies. Any unanticipated results should be discussed and explained. The results should be judged with respect to the modeling project objectives.

The results of any water resources modeling project typically include large files of time series data. Only the most dedicated of clients will want to read those files, so the data must be presented in a more concise form. Statistical summaries should explicitly include any restrictions and uncertainties in the results. They should identify any gaps in the domain knowledge, thus generating new research questions or identifying the need for more field observations and measurements.

Reporting Model Results

Although the results of a model should not be the sole basis for policy decisions, modelers have a responsibility to translate their model results into policy recommendations. Policymakers, managers, and indeed the participating stakeholders often want simple, clear and unambiguous answers to complex questions. The executive summary of a report will typically omit much of the scientifically justified discussion in its main body regarding, say, the uncertainties associated with some of the data. This executive summary is often the only part read by those responsible for making decisions. Therefore, the conclusions of the model study must not only be scientifically correct and complete, but also concisely formulated, free of jargon, and fully understandable by managers and policymakers. The report should provide a clear indication of the validity, usability and any restrictions of the model results. The use of visual aids, such as graphs and GIS, can be very helpful.

The final report should also include sufficient detail to allow others to reproduce the model study (including its results) and/or to proceed from the point where this study ended.

13.5.2.2 Evaluating Modeling Success

There are a number of ways one can judge the extent of success (or failure) in applying models and performing analyses in practice. Goeller (1988) suggested three measures as a basis for judging success:

1. How the analysis was performed and presented (analysis success).
2. How it was used or implemented in the planning and management processes (application success).
3. How the information derived from models and their application affected the system design or operation and the lives of those who use the system (outcome success).

It is often hard to judge the extent to which particular models, methods and styles of presentation are appropriate for the problem being addressed, the resources and time available for the study, and the institutional environment of the client. Review panels and publishing in peer-review journals are two ways of judging. No model or method is without its limitations. Two other obvious indications are the feelings that analysts have about their own work and, very importantly, the opinions the clients have about the analysts' work. Client satisfaction may not be an appropriate indicator if, for example, the clients are unhappy only because they are learning something they do not want to accept. Producing results primarily to reinforce a client's prior position or opinions might result in client satisfaction, but, most would agree, this is not an appropriate goal of modeling.

Application or implementation success implies that the methods and/or results developed in a study were seriously considered by those involved in the planning and management process. One should not, it seems to us, judge success or failure on the basis of whether or not any of the model results (the computer "printouts") were directly implemented. What one hopes for is that the information and understanding

resulting from model application helped define the important issues and identify possible solutions and their impacts. Did the modelling help influence the debate among stakeholders and decision-makers about what decisions to make or actions to take? The extent to which this occurs is the extent to which a modeling study will have achieved application or implementation success.

Outcome success is based on what happens to the problem situation once a decision largely influenced by the results of modeling has been made and implemented. The extent to which the information and understanding resulting from modeling helped solve the problems or resolve the issues, if it can be determined, is a measure of the extent of outcome success. It is clear that success in terms of the second or third criteria will depend heavily on the success of the preceding one(s). Modeling applications may be judged successful in terms of the first two measures but, perhaps because of unpredicted events, the problems being addressed may have become worse rather than improved, or while those particular problems were eliminated, their elimination may have caused other severe problems. All of us can think of examples where this has happened.

For example, any river restoration project involving the removal of engineering infrastructure is a clear indication of changing objectives or new knowledge. Who knows whether or not a broader systems study might have helped earlier planners, managers, and decision-makers foresee the adverse ecological consequences of converting rivers to canals, and whether or not anyone will care. Hindsight is always clearer than foresight. Some of what takes place in the world is completely unpredictable. We can be surprised now and then. Given this, it is not clear whether we should hold modelers or analysts, or even planners or managers, completely responsible for any lack of "outcome success" if unforeseen events that changed goals, or priorities or understanding did indeed take place.

Problem situations and criteria for judging the extent of success will change over time, of course. By the time one can evaluate the results, the system itself may have changed enough for the outcome to be quite different than what was predicted in the analysis. Monitoring the performance of any decision, whether or not based on a successfully analyzed and implemented modeling effort, is often neglected. But monitoring is very important if changes in system design, management and operation are to be made to adapt to changing and unforeseen conditions.

If the models, data, computer programs, documentation and know-how are successfully maintained, updated, and transferred to and used by the client institutions, there is a good chance that this methodology will be able to provide useful information relevant to the changes that are needed in system design, management, or operation. Until relatively recently, the successful transfer of models and their supporting technology has involved a considerable commitment of time and money for both the analysts and the potential users of the tools and techniques. It has been a slow process. Developments in interactive computer-based data-driven decision support systems that provide a more easily understood human-model-data-computer interface have substantially facilitated this technology transfer process, particularly among model users. These technology developments have had, and we think will continue to have, a major impact on the state of the practice in using models in support of water resources planning and management activities.

13.6 Conclusions

The effectiveness of strategies for dealing with issues of water quantity and quality, and their variability, has a major impact on the well-being of living species, and even the survival of some. How well water is managed also impacts the functioning and resilience of ecosystems, the vitality of

societies, and the strength and growth of economies. Fortunately we humans can determine which water resources development and management strategy will work best in a given situation, not only for the immediate future but in the long-run as well. And if conditions change, our strategies can adapt. To accomplish this we need to identify and evaluate the effectiveness of the water resources development and management alternatives available to us in an economic, hydrologic and sociopolitical environment that seems to be a constantly changing. We can do this through the use of various models, developing preferred strategies based in part on their results, and informed by the concerns and objectives of stakeholders and the decision making institutions.

This book has focused on ways of developing and using various optimization and simulation modeling methods for analyzing and evaluating water resource development and management alternatives. This final chapter has presented some guidelines for carrying out water resources planning projects, including its modeling components. Such projects are typically very complex and challenging.

Water management planning projects must address a complex and interconnected web of science, engineered infrastructure, legal regulations governing water use, societal expectations, and institutional structures and authorities that have evolved over time. Much of the current complexity that exists in various regions of the world has developed over time in response to changing interests and objectives of water users and environmental considerations. Although the impacts of changes in the climate on water supplies and demands are generally recognized,

these ongoing changes as well as the linkages between environmental and societal factors in specific basins and regions all lead to major uncertainties in the future.

The guidelines discussed in this chapter have been developed and used by Dutch experts in Deltares to assess water resources systems and to develop plans and strategies for managing them. Deltares has been actively involved in numerous water resources planning and management projects throughout the world. The approach described in this chapter illustrates how these projects are conducted, and the major factors that are considered while conducting them. The effects and impacts of some of their projects have been relatively local and required consideration of only a few sectors of the economy. Other, more comprehensive projects have had national or international impacts, and led to transboundary (international) compacts.

Clearly each water resources system is unique with respect to its management issues and problems and its institutional environment. Project planning and analysis approaches must adapt to these situations. Hence, each project will differ, and will no doubt need to deviate from the suggested guidelines presented in this chapter. Other approaches are possible and may be equally effective. What remains important in all cases is the establishment of a comprehensive, systematic process of planning and analysis together with constant communication among planners, decision-makers and the interested and affected public. The end result should be an improved, more sustainable, and equitable water resources development plan and management policy, appropriate for the region and its people.

Box 13.4. Example 1: Objectives and criteria adopted in West Java WRM study

Objectives	Evaluation criteria
<i>Socio-economic objectives and criteria</i>	
1. Improve employment (–)	Increase of employment by WRM strategies – Number of permanent jobs (#) – Number of temp. jobs (mn-year)
2. Increase income of people – Improve income position of farmers – Improve equity in income distribution	• Farmer net income (Rp/year) • Difference in benefits of WRM strategies per capita between: – Kabupaten (%) – Urban/rural areas (%) – Income groups (%)
3. Increase the non-oil export production (shrimps, tea, and rubber)	– Export value (Rp/year)
4. Support economic development in an economically efficient way	– Total annual. benefits (Rp/year) – Total annualized costs (Rp/year) – B/C ratio (–) – IRR (%) – NPV (Rp/year) – Total capital required (Rp) – Foreign currency required (%) – Total construction costs (Rp) – Total O&M costs (Rp) – Sectoral value added (Rp/year) – GRP (Rp/year)
<i>User-related (sectoral) objectives and criteria</i>	
1. Increase agricultural production (3% per year)	– Padi (ton/year) – Palawija (ton/year) – Export value of crops (or import substitution) (Rp/year) – Unit costs water supply (Rp/m³) – % failure meeting demand (%)
2. Increase power production (–)	– Installed capacity (MW) – Power production (GWh/year) – Failure meeting firm power (%) – Price of power prod. (Rp/Kwh) – Energy production value (Rp)
3. Increase fish production (–)	– Fish produced (ton/year) – Fish pond area (ha) – Export value (Rp/year)
4. Support industrial development • Water supply for industry (full supply) • Provision of opportunity for discharge of waste water	– Amount of supply (m³/s) – Cost of water supply (Rp/year) – Unit costs water supply (Rp/m³) – % failure meeting demand (%) – Cost to maintain water quality standards (Rp/year)
5. Enhance water-related recreation	
<i>Environmental and public health related objectives and criteria</i>	
1. Improve public health • Improve drinking water supply - urban: BNA, IKK and major city programs: 60 l/cap/day, serving 70 % - rural: 55 % • improve flushing (1 L/s/ha in urban area)	– Supply (1/day/ capital) – % of people connected (Rp/m³) – Price of drinking water (%) – % failure meeting demand (%) – Volume of flushing water (m³/s) – Unit costs (Rp/m³) – % failure meeting demand (%)

(continued)

Objectives	Evaluation criteria
2. Improve/conserv natural resources and environment • Erosion and sedimentation control (erosion <1 mm/year) • Conservation of nature • Water quality	– Area severely eroding (ha) – Erosion (mm/year) – Sediment yield (tons/year) – Reafforested area (ha) – Replanted area (ha) – Terraced area (ha) – % external wood supply to total wood demand (%) – Concentration water quality parameters (ppm)
3. Provide flood protection (return period: depending on value of endangered area)	- return period [years] - flood alleviation benefits (reduced damage) [Rp/year] - flood control cost [Rp/year] - number of people in endangered areas [#] - flooded area [ha]
<i>Planning and implementation related objectives and criteria</i>	
1. Take care of maximum agreement with existing policies in other fields of planning (e.g. economic regional planning)	– Deviations from/conflicts with existing policies
2. Maximize flexibility of proposed strategy	– Degree to which strategy can be adjusted to changes in demands, standards, technological innovations
3. Maximize reliability of proposed strategy	– Degree of certainty with which proposed strategy will meet the realization of objectives
4. Provide sufficient acceptance of proposed strategy by public, interest groups and executing authorities	– Degree of acceptance by parties involved
5. Takes care of maximum agreement of proposed strategy with existing competence and responsibilities of agencies concerned	– Deviations from/conflicts with existing competence and responsibilities

^aKabupaten = Indonesian administrative unit

^bPadi = Rice crop

^cPaliwija = Non-rice crop

^dRp = Rupiah

Box 13.5 Example 2: Score-card Egyptian National Water Resources Plan study

	Unit	1997 base	2017 reference case	Strategy facing the challenge
<i>General (middle scenario)</i>				
Population	Million	59.3	83.1	83.1
Urbanization	Ratio	0.44	0.48	0.48
GDP at economic growth of 6%	Billion LE	246	789	789
<i>Economic development objectives</i>				
Agriculture: irrigation area	Mfeddan	7.985	11.026	10.876
Gross production value	Billion LE	34.46	35.76	38.50
Crop intensity	Ratio	2.1	1.5	1.7
Net value production per feddan	LE/feddan	2812	2075	2153
Net value production per unit of water	LE/m ³	0.64	0.66	0.60
Export/import value	Ratio	0.09	0.12	0.20

(continued)

	Unit	1997 base	2017 reference case	Strategy facing the challenge
Industry: costs polluted intake water	LE/m ³	0.65–1.10	0.65–1.10	2.00
Wastewater treatment costs	LE/m ³	0.22–0.50	0.22–0.50	1.00
Fishery: production (index 100 in 1997)	Index	100	86	95
Tourism: navigation bottlenecks	Index	100	114	0
<i>Social objectives</i>				
Create living space in desert areas	% of tot. pop	1.5%	23%	22%
Employment and income Employment in agriculture	M pers. year	5.01	6.24	7.30
Employment in industry	M pers. year	2.18	4.99	4.99
Average income farmers	LE/year	5362	4629	4309
Drinking water supply Coverage	Percentage	97.3%	100%	100%
Sanitation Coverage	Percentage	28%	60%	60%
<i>Equity</i> Equity water distribution in agriculture	–, 0, +	0	+	+
Self-sufficiency in food: cereals	Percentage	73%	53%	46%
<i>Meeting water needs</i>				
Water resources development Available Nile water	BCM	55.8	55.5	55.5
Abstraction deep groundwater	BCM	0.71	3.96	3.96
Water use efficiency Nile system Outflow to sinks from Nile system	BCM	16.3	17.6	12.5
Overall water use efficiency Nile system	Percentage	70%	67%	77%
Water in agriculture Supply/demand ratio (1997 assumed 1.0)	Ratio	1.00	0.80	0.92
Water availability per feddan Nile system	m ³ /feddan/yr	4495	3285	3866
Public water supply UFW losses	Percentage	34%	34%	25%
Supply/demand ratio	Ratio	0.67	0.76	1.00
<i>Health and environment</i>				
Pollution and health <i>E. coli</i> standard violation (1997 = 100)	Index	100	121	110

(continued)

	Unit	1997 base	2017 reference case	Strategy facing the challenge
Water quality shallow groundwater	–, 0, +	0	–	–
Ecology and sustainability Sustainability: use of deep groundw.	Abstr/pot	0.15	1.00	1.00
Condition Bardawil (Ramsar site)	–, 0, +	+	–	+
Condition coastal lakes	–, 0, +	0	–	0

^aUFW = Unaccounted for water (the water that is lost in the system)

^bfeddan = 0.42 ha

^cLE = Egyptian pound

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